

Name:

Form:

MYP 1 Year 7 Super Skills

Statement of Inquiry:	Practical techniques and innovations allow us gather evidence of scientific relationships and construct models to help explain phenomena.
Key Concept:	Relationships
Related Concepts:	Models and Evidence



7SS MYP Command Terms

Term	Definition
Analyse	Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.
Apply	Use knowledge and understanding in response to a given situation or real circumstances
Describe	Give a detailed account or picture of a situation, event, pattern or process
Design	Produce a plan, simulation or model
Discuss	Offer a considered and balanced review that includes a range of arguments, factors or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence
Document	Credit sources of information used by referencing (or citing), following one recognized referencing system. References should be included in the text and also at the end of the piece of work in a reference list or bibliography
Evaluate	Make an appraisal by weighing up the strengths and limitations
Explain	Give a detailed account
Formulate	Express precisely and systematically the relevant concept(s) or argument(s)
Interpret	Use knowledge and understanding to recognize trends and draw conclusions from given information
Outline	Give a brief account
Present	Offer for display, observation, examination or consideration
Recall	Remember or recognize from prior learning experiences
Select	Choose from a list or group
Solve	Obtain the answer(s) using appropriate methods
State	Give a specific name, value or other brief answer without explanation or calculation
Suggest	Propose a solution, hypothesis or other possible answer
Summarize	Abstract a general theme or major point(s)

Glossary: Super Skills

Word	Meaning
Corrosive	A chemical that will attack your skin.
Flammable	A chemical that will catch fire easily
Toxic	A chemical that is poisonous.
Bunsen Burner	A piece of lab equipment used for heating by burning a gas.
diagram	A sketch or a picture of how you want to set up your equipment.
Beaker	Made out of glass. Used to investigate chemicals during practical work.
Filter Funnel	Used to remove a solid from a liquid.
Measuring cylinder	Used to measure out a liquid
Conical flask	A triangular shaped glass container.
Apparatus	Another name for the equipment that you use in the lab.
Thermometer or Temperature probe	Used to measure the temperature
Celsius	The unit for measuring temperature
Centimeter cubed (cm ³)	The unit for measuring the volume.
Stopwatch	Used for measuring time (in seconds)
Electronic balance	Used for measuring the mass of an object.
Particle	A very small part of something, they are drawn as being round in shape.
Theory	Used to explain why something works the way it does.
Solid	When all the particles are packed close together in a regular pattern.
Liquid	When all the particles are packed together but can move about in a random way.
Gas	When all the particles are far away from each other and move very quickly.
Property	Something we can see that needs explaining. E.g. Gases can be squeezed easily.
Expand/contract	Get bigger or smaller
Lighter	Used to produce a small flame to help light the Bunsen
Heat proof mat	Put under the Bunsen to stop the bench getting burnt.
Air hole	Near the bottom of a Bunsen, this hole can be moved to allow more air into it.
Tripod	Made of metal and has 3 legs. Used when you want to heat things up with your Bunsen.
Gauze	Made of a type of metal mesh. This is put on the tripod and the thing you want to heat sits on this.
Research Question	What you are trying to find out in your experiment?
values	The amounts of things you want to use in your experiment
Variables	Things that you can change that will change your results in an experiment.
Method, plan or design	What you plan to do in your experiment
Results	What happened in your experiment
Conclusion	What did you find out and why did this happen?
errors	What went wrong or was not accurate in your experiment?
Evaluation	How you think you could make your experiment better

7SS 1a Laboratory Rules

Failure to adhere to these rules will result in immediate sanctions.

1. Do not enter a laboratory unless a teacher is present.
2. Place bags and coats under the table or out of the way.
3. Listen carefully to safety instructions and read safety warnings on worksheets.
4. Note the safety symbols on bottles and containers of chemicals.
5. Always wear safety glasses during practical work when there is a risk to the eyes.
6. No eating or drinking in the lab.
7. Act sensibly in the laboratory and during practical work. Never run.
8. When Bunsen burners are used, long hair should be tied back, safety glasses should be worn and ventilation fans should be used.
9. Do not leave an experiment unattended.
10. In cases of broken glassware or spilled chemicals, notify the teacher immediately.
11. Chemical waste should be disposed of in an appropriately labeled container, not in the sink. This should be emptied by a technician after every lesson.
12. Wash hands after using chemicals. At the end of an experiment, clear away and rinse glassware and leave in the wash trays. Tidy away other equipment as instructed and make sure your work space is left clean and neat.
13. Do not take anything out of the laboratory without permission.
14. Appropriate footwear must be worn. No flip flops or open toed sandals.
15. IB Chemists must wear white coats during practical work.

I have read and understand these rules:

Student Signature _____

Parent Signature _____

School Nurse: 2765 4633
Medical Room: G020 (G/F)

7SS 1b Safety Symbols

Many things around a lab have special signs on them warning you of danger.



Danger

This is a general warning sign. It may be placed in an area where there is some broken glass or a spilt chemical. Or it may just remind you to be particularly careful when doing something or using particular pieces of equipment or chemicals.



Harmful or Irritant

This symbol warns that a chemical may harm your health if you do not use it properly. The word 'irritant' means that the chemical may give you a rash if you get it on you or make you choke if you breathe it in.



Corrosive

This symbol tells you that a chemical will attack your skin if you get it on you.



Highly flammable

This symbol warns you that a chemical will catch fire easily.



Risk of electric shock

This symbol means that if used wrongly, the equipment may give you a dangerous electric shock.



Toxic

This symbol means that a chemical is poisonous. Poisons can kill.



Biohazard

This symbol means that there is some type of living thing in an area which may make you ill.



Eye protection must be worn

This symbol reminds you to wear safety glasses or goggles when working in a certain area or using particular pieces of equipment or chemicals.



Wash your hands

This symbol reminds you to wash your hands after you have done an experiment.

Activity: Test yourself or a partner to see how many you can remember?

· Self-test: the nine symbols are Danger (general warning), Harmful or Irritant, Corrosive, Highly flammable, Risk of electric shock, Toxic, Biohazard, Eye protection must be worn, Wash your hands.

7SS 1c Safety with Chemicals

There are many chemicals in a lab that can harm you. All the most dangerous chemicals will have safety symbols on them. You should check to see what the dangers of a chemical are before you use it. The dangers can be found on the container or on a Safety Card.

Find out what the following safety symbols mean.



The four hazard symbols are: corrosive (attacks skin), toxic (poisonous), highly flammable (catches fire easily) and harmful/irritant (may harm your health). Check the Safety Card for each symbol.



Whenever you are using a chemical:



- DO NOT breathe it in.
- DO NOT let it get on your skin.
- DO NOT try to taste it.
- DO NOT have a lighted Bunsen burner anywhere near something that will catch fire.



- DO make sure you understand any safety symbols on the container.
- DO wear safety glasses or goggles.
- DO wash your hands after using it.
- DO tell your teacher if any of it gets spilt.

What safety symbols should be on containers of these chemicals?

- Hydrochloric acid Corrosive (and harmful/irritant) - hydrochloric acid attacks skin and eyes.
- Ethanol Highly flammable - ethanol vapour catches fire very easily.
- Bromine Toxic and corrosive - bromine is poisonous and attacks skin; its vapour is harmful.
- Copper sulphate Harmful/irritant - copper sulphate may harm your health if swallowed or if it touches your skin.
- hydrogen Highly flammable - hydrogen catches fire very easily and forms an explosive mixture with air.

Extension:

Use the internet to find out what the safety symbol on a bottle of mercury should be?

The toxic hazard symbol (skull and crossbones), because mercury and its vapour are poisonous.

A bottle of mercury should carry the TOXIC symbol (poisonous), plus the harmful to the environment symbol.

Why is mercury thought to be dangerous?

Mercury is a toxic heavy metal; its vapour is easily inhaled and can damage the nervous system and kidneys.

Mercury is a toxic heavy metal. Its vapour is poisonous, and breathing it in can damage the brain, kidneys and nervous system, so it is dangerous even at room temperature.

What symbol should be on a truck containing petrol/gasoline? Why?

The highly flammable (flame) symbol - petrol gives off vapours that catch fire very easily.

Highly flammable - petrol vapour catches fire very easily, so a spark or flame could ignite the whole tanker.

7SS 1d Safety Poster

You will usually find a set of 'Lab Rules' in a science lab. They are there for your safety!

The drawing below shows a lab where there are no safety rules.



- 1 Write down a list of all the things going wrong in this lab.

Eating and drinking, running, broken glass on the floor, unreported spills, bags in the walkway, loose long hair, no goggles, an unattended burning Bunsen, a test tube pointed at people.

- 2 For each of your answers, write a safety rule.

No eating or drinking; never run; report broken glass and spills at once; bags and coats under the bench; tie long hair back; wear goggles; never leave a Bunsen unattended.

- 3 Design a poster for your lab, showing one safety rule and the reason for having it.

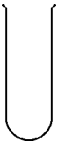
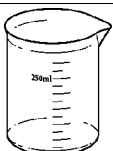
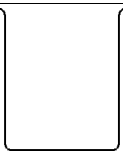
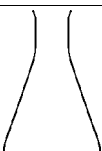
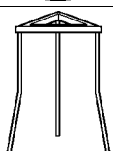
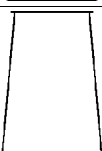
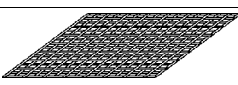
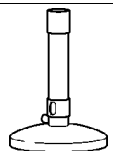
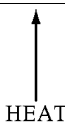
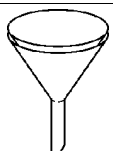
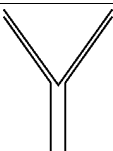
Poster: bold heading 'TIE YOUR HAIR BACK', a drawing of long hair catching fire from a Bunsen, the rule in large letters and the reason (hair can catch fire), plus a warning border.

7SS 2a Drawing Apparatus

Usually in science we draw **diagrams** of apparatus rather than pictures. Diagrams are easier to draw and make it easier to see how the apparatus is joined together.

Imagine that you could cut the apparatus in half. You draw what you would see when you look at the edge – this is called a **cross-section** diagram.

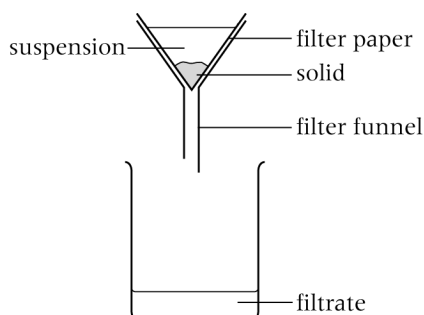
Here are some diagrams of apparatus that you might use.

Apparatus	Name	Diagram	What it is used for
	Test tube		Storing or mixing solids and liquids
	Boiling tube		Heating solids and liquids
	Beaker		Holding liquids or solids
	Conical flask (i.e. cone-shaped)		Holding and mixing liquids
	Tripod		Heating a beaker, flask or crucible over a Bunsen burner
	Gauze		Supporting a beaker or flask and spreading the heat from the flame
	Bunsen burner		Heating things
	Evaporating basin		Evaporating the water from a solution
	Filter funnel (with paper)		Separating an insoluble solid from a liquid
	Rubber bung		Keeping things in a tube or flask

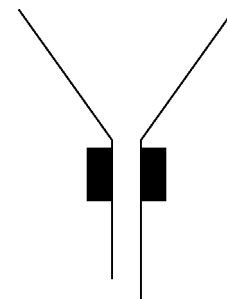
7SS 2b Drawing Assembled Apparatus

We use diagrams in science to show how pieces of apparatus are put together to do practical work.

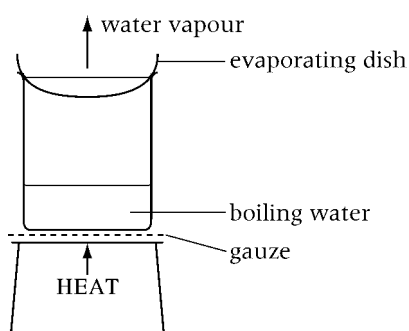
Example 1: Filtering



You don't usually need to show the clamps and stands. You know that the filter funnel is not really floating in mid-air. Clamps can be shown like this:

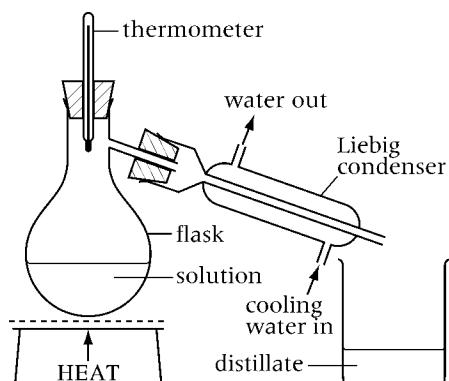


Example 2: Evaporating



Keep it simple! You can use an arrow to show heat instead of wasting time trying to draw a Bunsen burner.

Example 3: Distillation



This is a much more complicated diagram. Look carefully at the bungs. See how the diagram shows that the thermometer and tubes *go through* the middle of the bungs. Be careful that you don't 'block off' tubes that are really open. See how the 'water in' and 'water out' tubes are left open at the ends.

Make your own neat copies of one or more of the diagrams on this sheet. Don't forget to use a pencil and a ruler. Also make sure you label your diagram.

Extension: Find out and explain how you think distillation works. What sorts of mixtures can be separated using distillation?

The liquid with the lowest boiling point boils off first; its vapour is cooled in the condenser and collected as a liquid. It separates liquids with different boiling points, e.g. ethanol and water.

7SS 2c Making Measurements

Aim: How to read scales and accurately take measurements and use experimental equipment.

Materials required

Thermometers, beakers and measuring cylinders of various sizes, kettle, 3 burettes set up as demo. Stopwatch, meter ruler, tennis ball, ice.



Safety.

Treat all chemicals as irritants.
Wear goggles.

Method.

A range of activities are set up on the side benches. You can do the activities in any order.

1. Taking temperatures readings.

There are three beakers filled with warm/cold liquids. Use the thermometers to accurately record the temperatures.

Temp A = 65 °C B = 30 °C C = 8 °C

2. Taking volumes readings.

There are 2 different sized measuring cylinders containing liquids. Take a reading of the volume of each.

Volume D = 50 cm³ E = 25 cm³

There is also a measuring cylinder inverted in a bowl of water, containing a gas (F) and a liquid set up in a burette (G). Read and record these volumes as well.

Volume F = 18 cm³ G = 24.5 cm³

3. Taking times and distance readings.

Use the meter rulers to measure 1m height and 2 m in height. Be careful if you need to stand on a stool to do this. Use a tennis ball to record the time taken for the ball to hit the ground at 1m (H) and 2m (I).

Time at 1m (H) = 0.45 secs Time at 2m (I) = 0.64 secs.

4. Measuring Mass.

Use an electronic balance to accurately record the mass of a small conical flask (J).

Mass of conical flask (J) = 85.6 g

Analysis/Conclusion:

1. Which piece of equipment is most precise for measuring volumes? Beakers, measuring cylinders or burettes? Why?

A burette is most precise: its smallest scale division (0.1 cm³) lets you read the volume most exactly.

Burettes, because they have the finest scale (0.1 cm³ or smaller divisions) and a narrow tube, so small changes in volume can be read accurately. Beakers are the least precise.

Burettes, because they have the smallest scale divisions, so the uncertainty in each reading is smallest.

2. How can you make your measurements more accurate?

Repeat readings and take a mean, read at eye level to avoid parallax, and use the most precise instrument.

Read the scale at eye level with the bottom of the meniscus, use the most precise instrument, repeat readings and take an average, and keep the equipment and conditions the same each time.

3. Why don't we use beakers to measure out volumes?

Beakers have only rough graduations and a wide mouth, so they give rough, approximate volumes only.

Beaker markings are only rough approximations and a beaker is wide, so the same volume gives a very different height - this causes a large error, so beakers are not used for measuring volumes.

4. Which is most precise for measuring out 15cm³ of acid? A 20 cm³ measuring cylinder or a 50cm³ measuring cylinder? Why?

The 20 cm³ measuring cylinder, because its scale divisions are closer together (finer) than a 50 cm³ cylinder's, so 15 cm³ is read with a much smaller uncertainty.

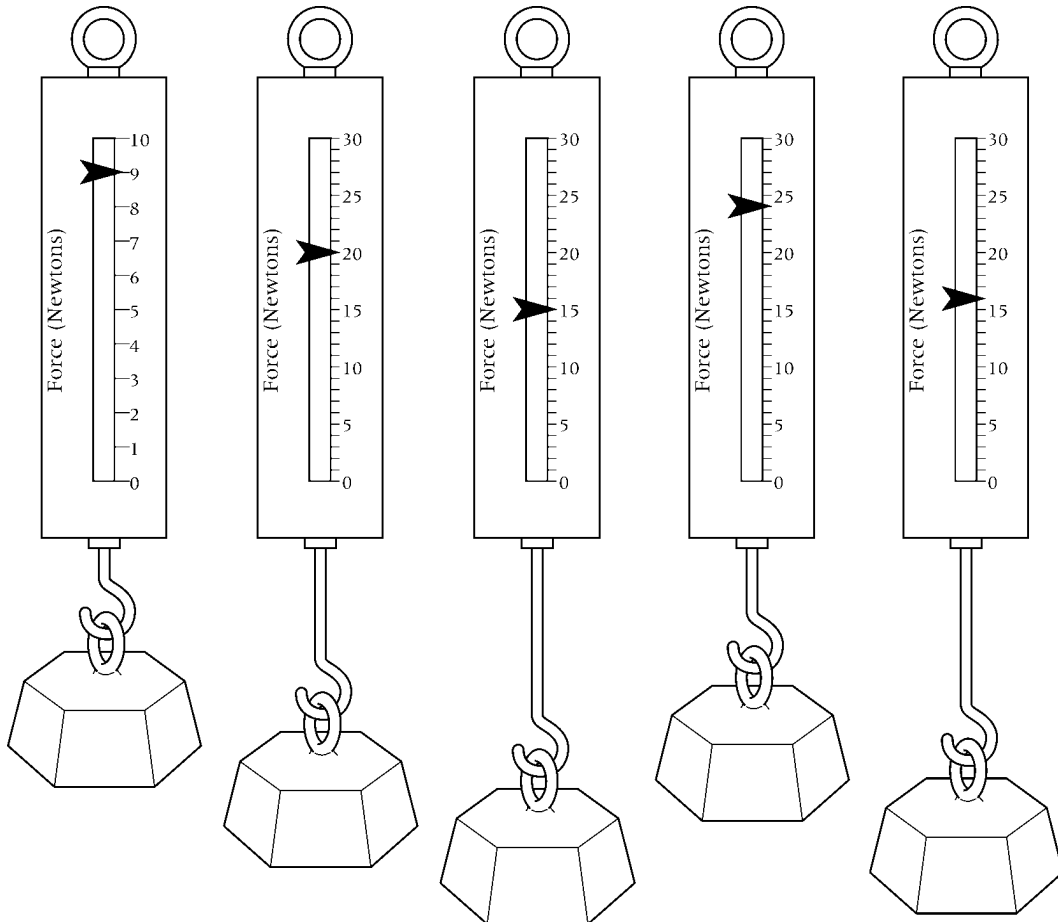
Extension: Find out what a pipette is. What it is used for? How is it used?

A pipette is a narrow glass tube that delivers one exact fixed volume (e.g. 10 cm³) of liquid. It is filled with a pipette filler (never your mouth) up to the mark, then the liquid runs out.

7SS 2d Reading Scales 1

When you investigate things in science, you usually need to measure something. Being able to read numbers from **scales** on **measuring instruments** is very important.

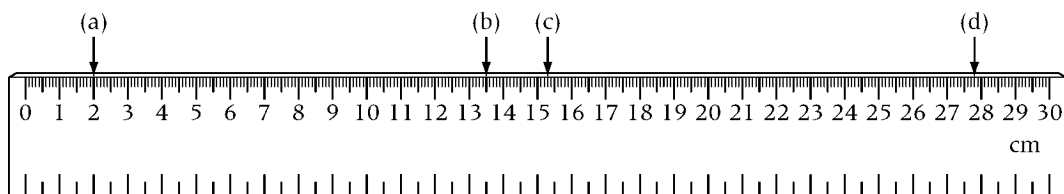
1 How much force are these force meters measuring? Remember the units.



Read the force meter where the pointer sits and give the value with its unit - the readings are in newtons (N), read to the nearest 0.1 N, e.g. 2.0 N.

9.2 N	20 N	15 N	25 N	15 N
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2 What are the readings shown by the arrows? Don't forget the units!

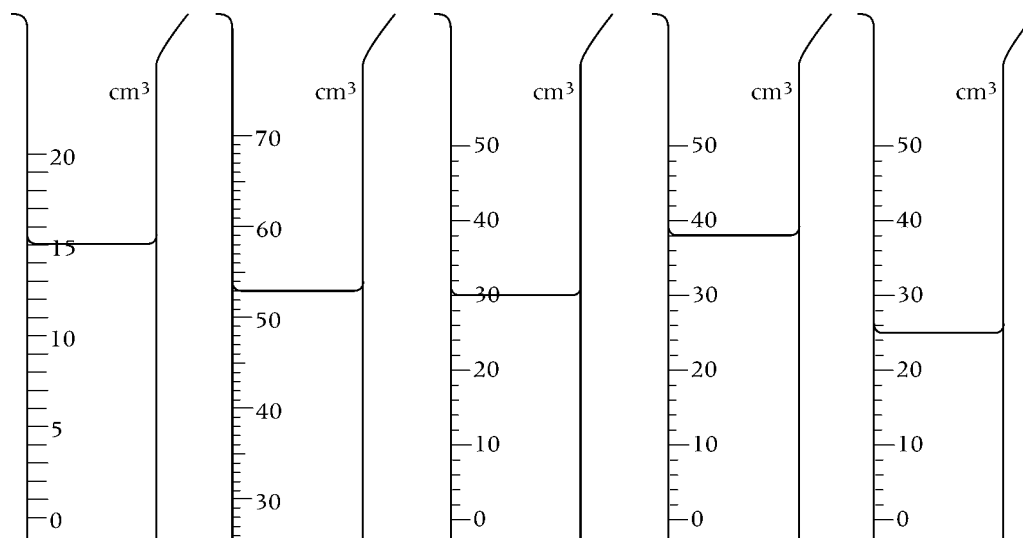
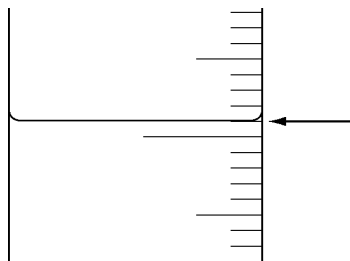


(a)	(b)	(c)	(d)
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Read each scale where the arrow points and give the value with its unit (cm, m, N, g or s), taking the reading to the nearest small division.

7SS 2e Reading Scales 2

3. How much liquid is in these measuring cylinders? Remember to read from the bottom of the meniscus, like this: The unit is cm^3 (This is the same as ml)



Read each measuring cylinder at the bottom of the meniscus with your eye level with the mark, and give the volume in cm^3 (ml), e.g.

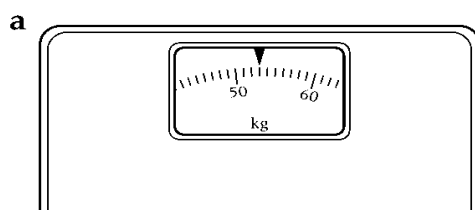
10 cm^3 , 24 cm^3 , 37 cm^3 , 15.5 cm^3 , 9 cm^3

meniscus at eye level -
value in cm^3 (ml)

meniscus at eye level -
value in cm^3 (ml)

Read the bottom of the
meniscus at eye level -
value in cm^3 (ml)

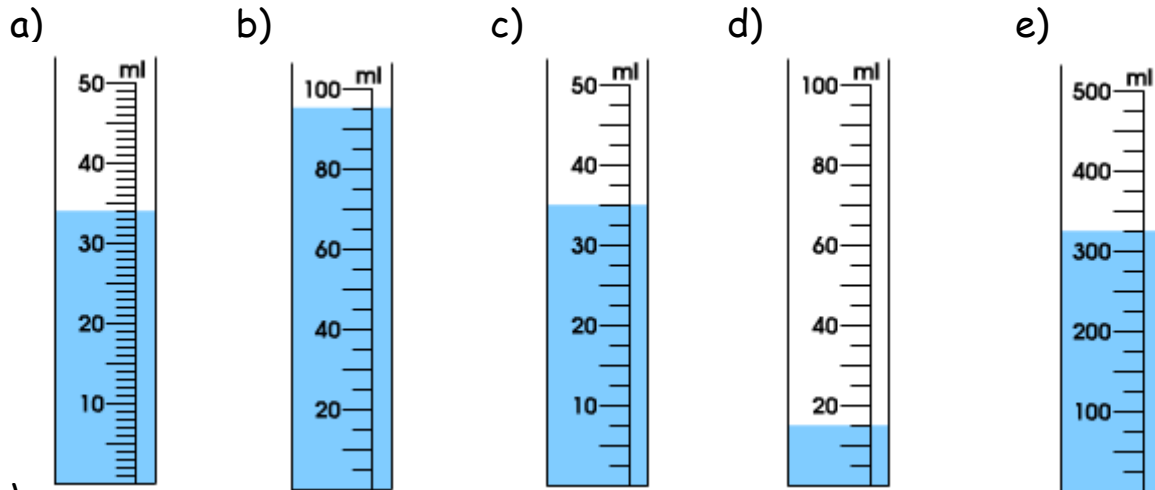
4. What are the following balances reading? Don't forget the units (kg).



Read the balance where the pointer or display shows, and give the mass in kilograms (kg) to the nearest division, e.g. 0.250 kg.

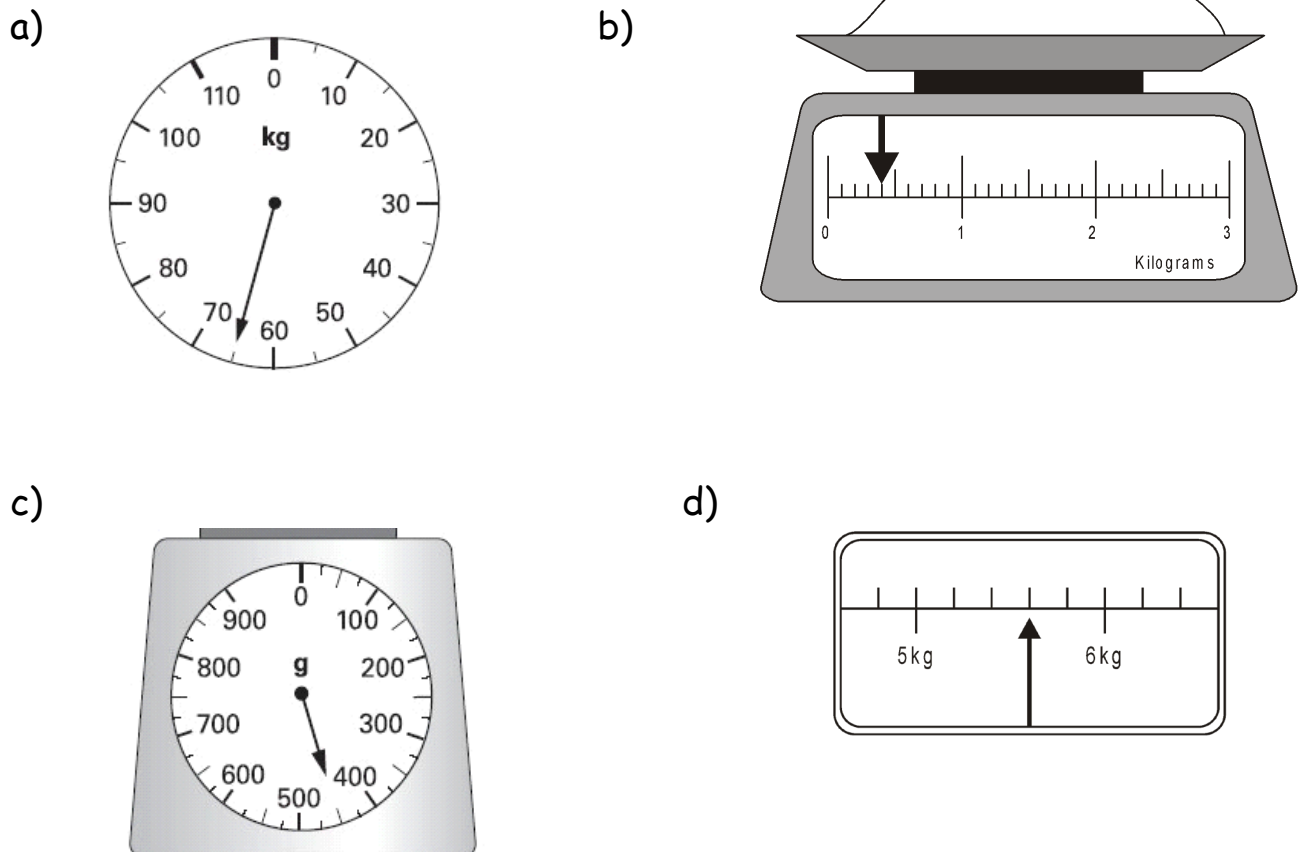
7SS 2f Reading Scales 3

Volume



Which is the most accurate scale?

Mass



The most accurate scale is the one with the smallest divisions (the finest scale), because it can detect the smallest change in mass, e.g. a digital balance reading to 0.01 g.

Read each scale where the pointer or display shows and record the mass in grams (g) to the nearest division of the scale.

Temperature

a)



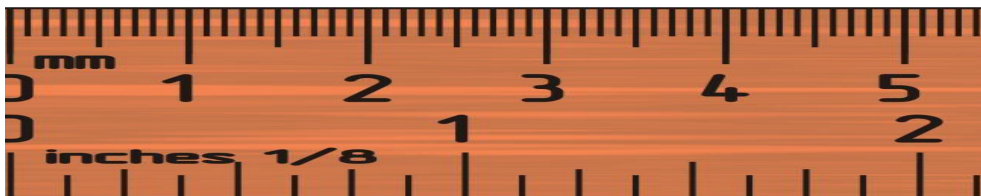
b)



Length

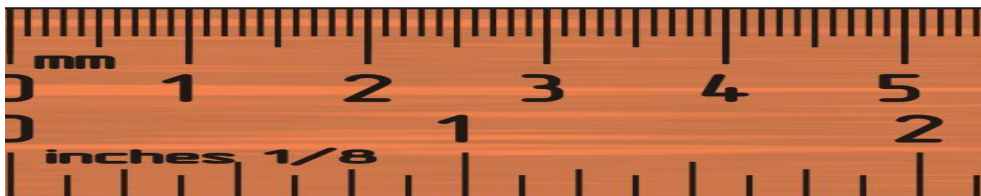
a)

Read each thermometer at eye level and record the temperature in °C to the nearest division, e.g. 25 °C.



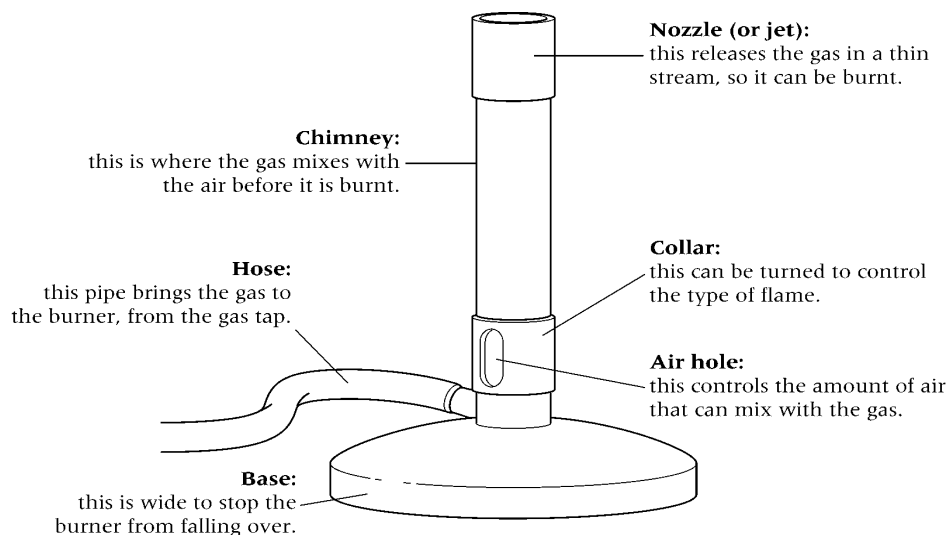
b)

Place the ruler along the object, keep your eye directly above the mark, and record the length in cm (or mm), starting from the 0 mark of the ruler.



7SS 3a The Bunsen Burner

The Bunsen burner is used to heat things in the laboratory. The diagram below shows the different parts of the Bunsen burner and explains what each part does.



Remember that a Bunsen burner will get hot when it is used, so never take one apart unless you are told to do so by your teacher. Always let a Bunsen burner cool down before you put it away.



Safety.

Wear Goggles. Tie long hair back.

Label the Bunsen: chimney (barrel), collar with the air hole, base, gas hose and gas tap. The air hole lets air mix with the gas; turning the collar opens or closes it.

How to light a Bunsen burner

Every time you use a Bunsen burner, you should follow these steps:

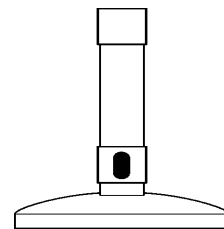
- 1 Check the hose for breaks and holes. DO NOT use the Bunsen burner if you find any breaks or holes. Give it to your teacher.
 - 2 Connect the hose to the gas tap but do not turn it on yet.
 - 3 Check that the air-hole in the collar of the Bunsen burner is **CLOSED**.
 - 4 Light a match or use a safety lighter and hold it a little distance (about 2 cm) above the top of the chimney of the Bunsen burner.
 - 5 Turn on the gas at the gas tap.
 - 6 The Bunsen burner will now light and give you a purple flame.
 - 7 Turn down the gas supply at the gas tap until you have the size of flame needed for your experiment.
 - 8 IF YOUR BUNSEN BURNER GOES OUT, TURN OFF THE GAS AT THE GAS TAP STRAIGHT AWAY.
- (If you wish, you can take a photo or use your phone to make a short video clip to remind you of how to light a Bunsen).

Types of Bunsen burner flames

The Bunsen burner will give **three** different types of flame:

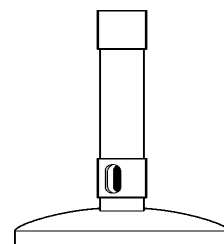
Roaring flame:

- air hole open, so lots of air mixes with the gas
- noisy, blue flame
- very hot flame
- used for heating things quickly



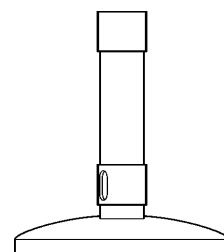
Medium flame:

- air hole half-open, so some air mixes with the gas
- light blue/purple flame, which is quieter than the roaring flame
- flame is quite hot
- used for heating liquids, especially if you are using a boiling tube



Safety flame:

- air hole closed, so hardly any air mixes with the gas
- quiet, purple, wavy flame
- Flame is not as hot as the medium flame
- not used for heating, because the flame leaves a layer of soot on things



When you have practiced lighting your Bunsen burner, complete the table below and draw the three flames you have seen. Use this sheet to help you.

Type of flame	Safety	Medium	Roaring
Type of air hole	CLOSED	HALF-OPEN HALF-OPEN	OPEN
Amount of air mixing with gas	None None	Some	Lots
Amount of noise	Very quiet	Fairly quiet (quieter than roaring)	Noisy / loud
Main colour of flame	Purple (wavy)	Light blue/purple	Blue
Diagram, sketch or photo of flame	Sketch: tall, wavy purple flame with a soft glowing tip (safety flame).	Sketch: steady light blue-purple flame, shorter than the safety flame, with a visible inner cone (medium).	Sketch: short, sharp blue flame with a paler blue inner cone and a roaring jet (roaring).
Amount of heat	Least Least	Quite hot	Hottest

Extension: Explain why opening the air hole produces the hottest flame. Use the internet to help if needed.

Opening the air hole lets more air (oxygen) mix with the gas, so the gas burns more completely. Complete combustion releases more energy, so the flame is hotter and appears blue.

7SS 3b Investigating which Bunsen Burner Flame is Hottest

Aim: How to safely use a Bunsen. How to plan and carry out investigations.

Materials required

Thermometers, beakers, measuring cylinders, tripods and gauzes, stop watches.



Safety. Wear goggles, tie long hair back, the equipment will get hot, let it cool before packing away.

Pour 100 cm³ of water into a beaker on a gauze and tripod; record the start temperature. Heat for 2 min with the air hole closed, then half-open, then open. Record each final temperature and repeat.

Plan. (If appropriate)

1. Plan an experiment to try and find out which of the three possible Bunsen flames are the hottest. You may do this on your own or in a group, or your teacher may give you some instructions/help.
2. What things will you need to measure?
3. What things will you need to change?
4. What things will you need to keep the same?
5. How will you set up your equipment? Draw a diagram.
6. What will you need to do? Write a method as a numbered list like a recipe.

Put 100 cm³ of water in a beaker on a gauze and tripod, note the starting temperature, heat 2 min with each flame (safety, medium, roaring) and measure the rise. Biggest rise = hottest.

Carrying out

1. Carry out your plan safely.
2. If you haven't made a plan, your teacher will give you some instructions.
3. Record your results in a table. Your teacher will help you to draw this. Remember to include titles and units in the columns.

The temperature of the water in °C at the start and at the end of heating, and the time in seconds - so I can calculate the temperature rise.

The type of Bunsen flame - safety (air hole closed), medium (half-open) and roaring (open) - changed by turning the collar.

Analysis/Conclusion:

1. If you have time, draw a bar graph of your results. Your teacher will help you do this and there is some help on the extension sheets at the back of your booklet.
2. Which of the Bunsen flames was the hottest? How do you know?
3. Why was this a fair test?
4. How could you have improved this investigation or made the results more accurate?

Extension: Look closely at the hottest Bunsen flame, where is the flame the hottest? Sketch and label a Bunsen. Find out the chemical formula of the gas and the equation which shows it burning in the air.

Why should you not try to heat something that is put really close to where the gas comes out of the top of the Bunsen?

Same volume of water (100 cm³), same starting temperature, same heating time (2 min), same beaker/tripod/gauze/thermometer, same flame distance and gas tap setting.



MYP Science

Practical Report Template

Name:

Form:

UNIT:	Super Skills - practical techniques and investigations
Statement of Inquiry:	Practical techniques and innovations allow us gather evidence of scientific relationships and construct models to help explain phenomena.
Key Concept:	Relationships
Related Concepts:	Models and Evidence
Investigation Title:	Investigating which Bunsen burner flame is the hottest

Research Question

(What are you trying to find out?)

How does the type of Bunsen burner flame (safety, medium or roaring) affect the temperature rise of 100 cm³ of water heated for 2 minutes?

Background Scientific Information and Hypothesis

(What do you think will happen and why?)

A Bunsen burns methane. The air hole controls the oxygen mixing with the gas: closed = safety flame (least heat), half-open = medium, open = roaring (blue, hottest). More oxygen = complete combustion.

I predict the roaring flame will heat the water the most, because with the air hole open more oxygen mixes with the gas so it burns more completely and releases more energy.

Variables

Independent <i>(What will change and how will you change it?)</i>	The type of Bunsen flame: closed air hole (safety), half-open (medium) and fully open (roaring), changed by turning the collar.
Dependent <i>(What will you measure to gather result?)</i>	The temperature rise of the water in $^{\circ}\text{C}$, measured with a thermometer (finishing temperature - starting temperature) after heating for 2 minutes.
Control <i>(What variables will you keep the same to make it a fair test? How will you keep the variables constant?)</i>	Volume of water (100 cm^3), starting water temperature, heating time (2 min), same beaker, tripod, gauze and thermometer, same flame distance and gas tap setting.

Apparatus Set-up

(Draw a labelled diagram of the assembled apparatus that you will use. Write the name of any equipment not shown in the diagram)

· Diagram: beaker of water on a gauze and tripod, Bunsen underneath, thermometer in the water. Label Bunsen burner, air hole, collar, tripod, gauze, beaker, water, thermometer, stopwatch.

Method

(Write a step by step account of how you conduct your experiment – the method must be DETAILED)

1.

2.

3.

4.

5.

1. Pour 100 cm³ of water into a beaker and stand it on the gauze and tripod.
2. Check the Bunsen burner for holes, close the air hole, light the burner and record the starting temperature of the water.
3. Heat the water for 2 minutes with the safety flame and record the finishing temperature.
4. Turn off the gas and let the equipment cool. Repeat with the air hole half-open (medium) and then fully open (roaring), using fresh water at the same starting temperature each time.
5. Calculate the temperature difference for each flame, repeat each flame three times and work out an average.

Safety

(How will you make sure your method is safe?)

Wear safety goggles and tie long hair back; keep bags and coats out of the way; clear flammable items from the bench; never leave the burner unattended; let the equipment cool before packing away.

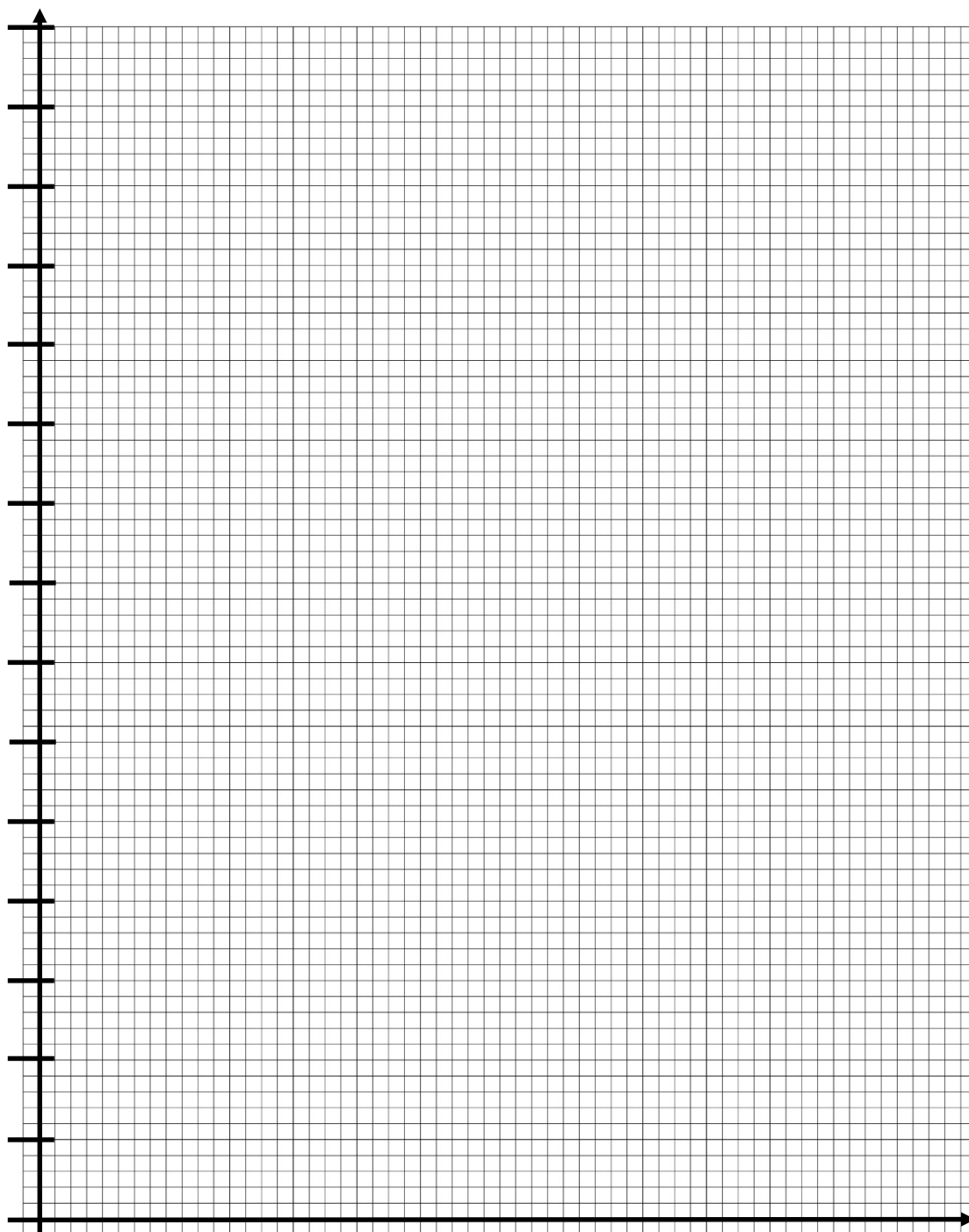
Reliability

(What will you do to ensure your results are reliable?)

Repeat each flame three times and use the average, use the same equipment and volume of water each time, read the thermometer at eye level, and repeat any anomalous reading.

7SS 3c Results table and graph template

Type of Flame	Starting Temperature (°C)	Finishing Temperature (°C)	Temperature difference (°C)
Safety	20		18
Medium	20		45
Roaring	20		60



Results

Raw Data

(A neat, organized table of results should be drawn, with headings, units, repeats and averages include)

Table headings: Type of flame | Starting temperature (0C) | Finishing temperature (0C) | Temperature difference (0C), with rows for safety, medium and roaring, 3 repeats and an average.

Observations

(What changes did you notice as the experiment was conducted, e.g. colour change, bubbles, heat change?)

The water started cold and clear. Bubbles formed on the bottom and sides of the beaker and rose as it heated; steam appeared with the hotter flames; the gauze glowed and the beaker became hot.

Graphing Results

You should draw a graph of your results and attach it to this report.

Analysis of Graph

(What did your results show? Are there any trends or patterns in your results? Can you give an example of a pattern or trend from your results?)

The temperature difference increases from the safety flame to the medium flame and is largest for the roaring flame - the roaring bar is tallest, so the hotter the flame the bigger the rise.

Conclusion and Evaluation

Scientific Conclusion and Explanation of Results

(Summarize what your results show. Does this support your prediction? Can you explain your results using scientific knowledge?)

My results show the roaring flame heated the water most, so it is the hottest - this supports my hypothesis. More oxygen mixes with the gas when the air hole is open, so it burns completely.

Evaluation of Scientific Method and Results

How reliable were your results? How well did your investigation answer the research question? What went well with your method?)

My results were fairly reliable because each flame was repeated three times and the repeats were close; the investigation answered the research question, showing the roaring flame was hottest.

(Were there any problems or parts of your investigation that were not very accurate or could have been done better?)

Issue with justification	Suggested SCIENTIFIC Improvement
Issue: heat was lost from the beaker to the air, so the temperature rise measured was smaller than the true rise (not all the energy went into the water).	Improvement: insulate the beaker (lid or cotton wool) and start with the water at the same temperature so less energy escapes as heat.
Issue: the thermometer had 1 °C divisions, so temperatures could only be read to the nearest degree and small changes were missed.	Improvement: use a digital thermometer reading to 0.1 °C and read it at eye level to reduce reading error.
Issue: the Bunsen was not always exactly the same distance below the gauze, so the flame did not heat the water identically each time.	Improvement: clamp the Bunsen and measure a fixed distance (e.g. 5 cm below the gauze) with a ruler before each trial.
Issue: some flames were only tested once, so an anomalous reading would change the conclusion.	Improvement: repeat each flame three times, calculate an average, identify anomalies and re-test them.

MYP Practical Report Checklist - Student

Before you submit your practical report to your teacher you must use the checklist below to ensure all parts of your report have been completed.

	Section	What you need to do	Completed Y/N
DESIGN (MYP Criterion B: Inquiring and Designing)	Title	Give a name to the experiment you are conducting	
	Research Question	The research question should be clear, concise , and include the independent variable and the dependent variable.	
	Background Information	You should give the science theory behind your investigation, and it should include scientific terminology. This should be presented as a short scientific paragraph .	
	Hypothesis	What do you predict will happen in your experiment and why ? Yes - all tables, axes and straight lines are drawn in sharp pencil using a ruler.	
	Variables	Independent Variable - what will you change and how will you change it? Dependent Variable - what will you measure and how will you measure it? Control Variables - what are they? How and why are they controlled? You change only this one thing between runs (e.g. different drop heights) and keep every other variable the same.	
	Apparatus	Have you completed a diagram of assembled equipment that shows how to successfully collect data? Have you used a ruler and pencil to do this? Have you labeled all the chemicals and equipment in your diagram? Yes - label every part of the diagram, include names and units, and give the diagram a title. Measure it with a suitable instrument (ruler, balance, stopwatch), read the scale carefully and repeat for accuracy.	
	Method	Have you written your method in the following format: - Stepwise - Logical order of steps in investigation - Used correct names for chemical and equipment - Provided detailed account of the steps in the method - Describe how you will change the independent variable - Describe how control variables are monitored Keep them the same in every run (same surface, equipment, start point and temperature) so the test is fair.	
	Safety	Discuss any safety issues with chemicals or equipment being used, and describe what precautions should be taken to ensure you conduct the investigation safely.	

RESULTS and CONCLUSION (MYP Criterion C: Processing and Evaluating)	Results Table	A table of results must include: - Headings - Units - Repeats - Averages	
	Observations	Record detailed observations: - At start of experiment - During experiment - At end of experiment Observations can include colour change, bubbles, heat etc.	
	Calculations	Calculate averages of repeats. Show formula for calculation Show working out for calculations Give final answer to calculation with correct units .	
	Graph	Draw a graph to include the following; - Heading, labels with units on each axis - Suitable scales that occupy minimum $\frac{3}{4}$ of each axis - Plot all points accurately - Draw a curve or straight line of best fit that ignores points that do not fit the pattern. - Discuss trends or patterns observed in graph.	
	Conclusion	Summarize what your results show. Does this support your prediction/hypothesis ? Explain your results using scientific knowledge ?	
	Evaluation	Discuss the reliability of your results . Discuss the validity of the method you used. Say whether the results match the prediction and quote the data, e.g. 'bounce height doubled when drop height doubled, so yes'.	
	Improvements	Suggest improvements to the scientific method and describe how they can reduce or eliminate errors	

Use scientific knowledge/equations to explain the pattern, e.g.
a higher drop gives more gravitational potential energy to transfer.

7SS 5a Constructing Table of Results

A table is a way of showing a lot of information in a way that is easy to read.

This ...

Bike	Bike	Number of pupils
Ordinary bike	16	16
Mountain bike	9	9
Racing bike	5	5
No bike	4	4

is easier to read than this ...

Out of all the children in the class, 16 of them have an ordinary bicycle. Four of the pupils did not have a bike at all. Racing bikes were owned by five pupils, and the other nine all had mountain bikes.

Scientific Results Tables

When you do a practical, you often need to write your results in a table. You need to draw your table before you start your experiment. This table is for an experiment to find out how long it takes to dissolve different amounts of sugar in some hot water.

Amount of sugar (spatulas)	Time to dissolve (minutes)
1	
2	
3	
4	
5	

This is what you are changing in the experiment, so it always goes in the first column.

This is what you will be measuring.

Always put the units in.

These are the amounts of sugar you will be testing.

Use a ruler when you draw straight lines.

Sometimes you investigate more than one thing. You might want to find out how long it takes the sugar to dissolve in hot water and in cold water. Your results table for this experiment should look like this:

Amount of sugar (spatulas)	Time to dissolve (minutes)	
	Hot water	Cold water
1		
2		
3		
4		
5		

This heading applies to both the columns underneath it.

Extension: Construct a neat results table to find out; what is the favourite flavour of chips/crisps in your class?

Try and organise this information to make it clear and easy to understand, just like the examples above.

Two-column table: 'Flavour of crisps' and 'Number of pupils'. One row per flavour (e.g. Ready Salted, Cheese & Onion, Salt & Vinegar, Prawn Cocktail, Other) plus a Total row.

7SS 6a Bar Charts

Do you need a bar chart or a line graph?

The easy way to decide which kind of graph to draw is:

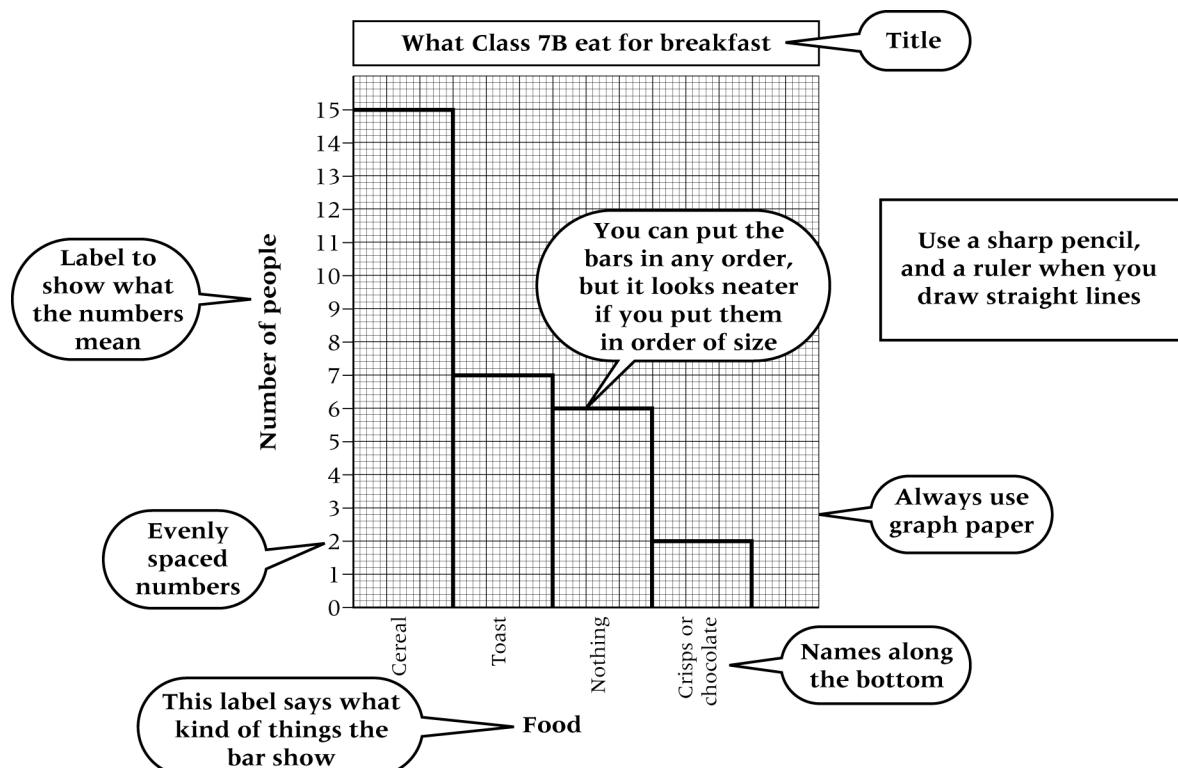
- If you have **measured two** things (e.g. time, length or temperature) you need a **line graph**.
- If you have counted something (e.g. how many insects live in different places) you need a **bar chart**.
- If one of the things you have studied can be described in words (e.g. different kind of food) you need a **bar chart**.

Breakfast

You have done a survey to show what the people in your class had for breakfast, and these are your results:

Food	Number of people
cereal	15
toast	7
crisps or chocolate	2
nothing	6

You would draw a bar chart like this one to show your results:



A line graph if the independent variable is continuous numbers; a bar chart if it is categoric (named groups).

- 1 Draw a neat version of the bar chart above. Use graph paper and use a ruler when you draw straight lines. (graph paper on next page)
- 2 Draw bar charts to show the results of these surveys.

a Number of plants in a lawn

Bar chart: title 'Number of plants in a lawn'; x-axis 'Plant' (Daisy, Buttercup, Clover, Thistle); y-axis 'Number of plants' 0-20; bars 10, 7, 20 and 2; scale 1 square = 2 plants.

Plant	Plant	Number	Number
Daisy	Daisy	10	10
Buttercup	Buttercup	7	7
Clover	Clover	20	20
Thistle	Thistle	2	2

b Hair colour in a class

Bar chart: title 'Hair colour in a class'; x-axis 'Colour' (Black, Brown, Blonde, Red); y-axis 'Number of people' 0-12; bars 12, 10, 8 and 1; scale 1 square = 2 people.

Colour	Colour	Number of people	Number of people
Black	Black	12	12
Brown	Brown	10	10
Blonde	Blonde	8	8
Red	Red	1	1

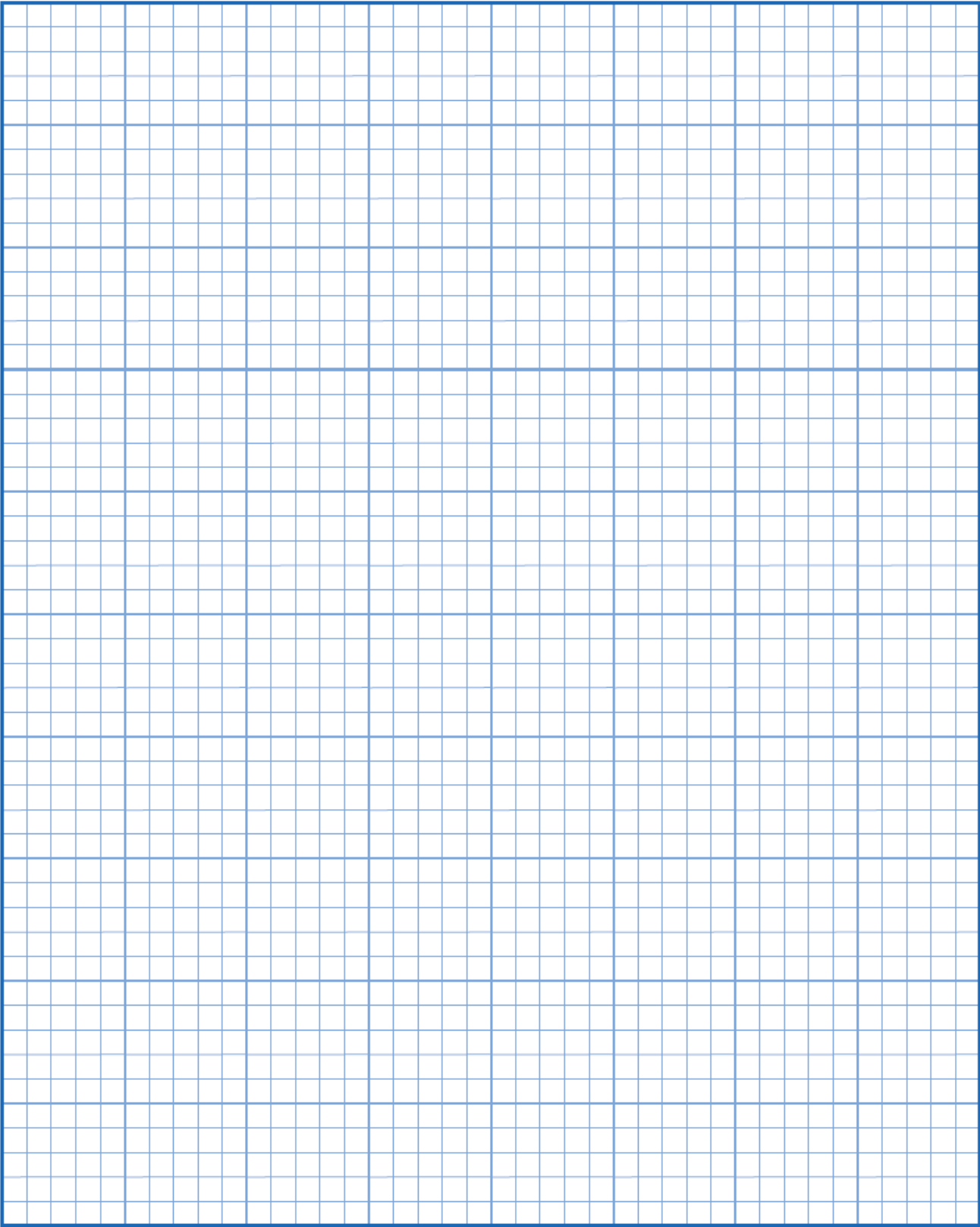
- 3 Carry out your own survey to find out the following information and draw bar charts to show your results.

a How the pupils in your class get to school.

b How many brothers and sisters' pupils in your class have.

Survey: tally how each pupil gets to school (walk, bike, car, bus), then bar chart with 'Way of getting to school' on the x-axis and 'Number of pupils' on the y-axis.

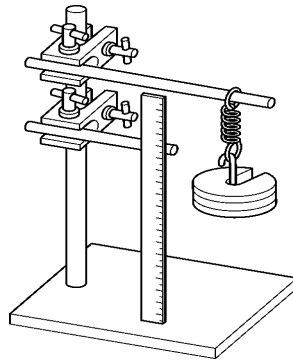
Survey: tally how many brothers and sisters each pupil has (0, 1, 2, 3, 4+), then bar chart with 'Number of brothers and sisters' on the x-axis and 'Number of pupils' on the y-axis.



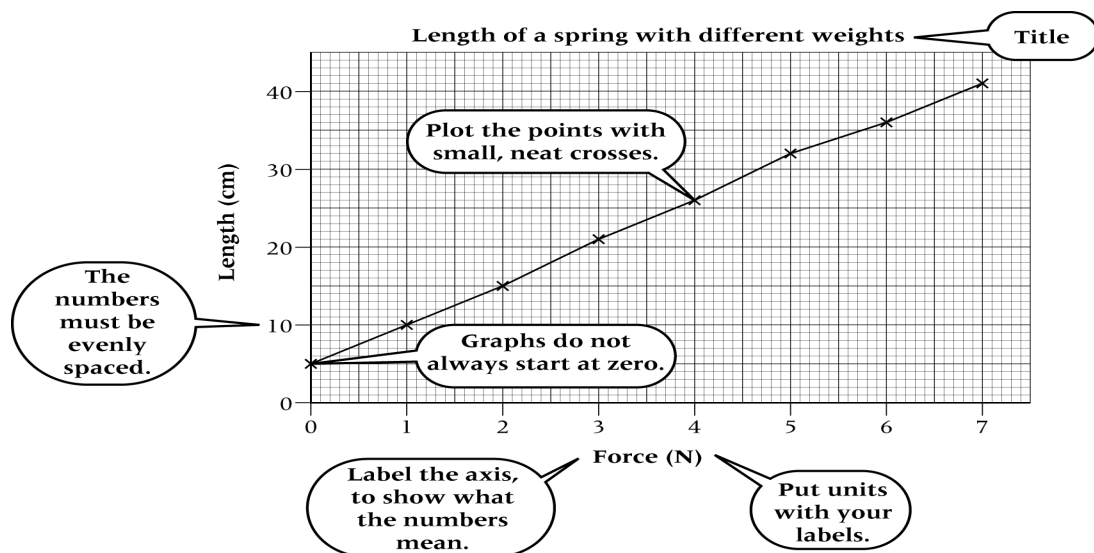
7SS 6b Line Graphs

Line graphs can be useful for showing the results of an experiment. You usually use a line graph when you have to show two sets of measurements.

Here are the results of an experiment to find out how long a spring is when it has different forces on it.



Force (N)	Length (cm)
0	5
1	10
2	15
3	21
4	26
5	32
6	36
7	41

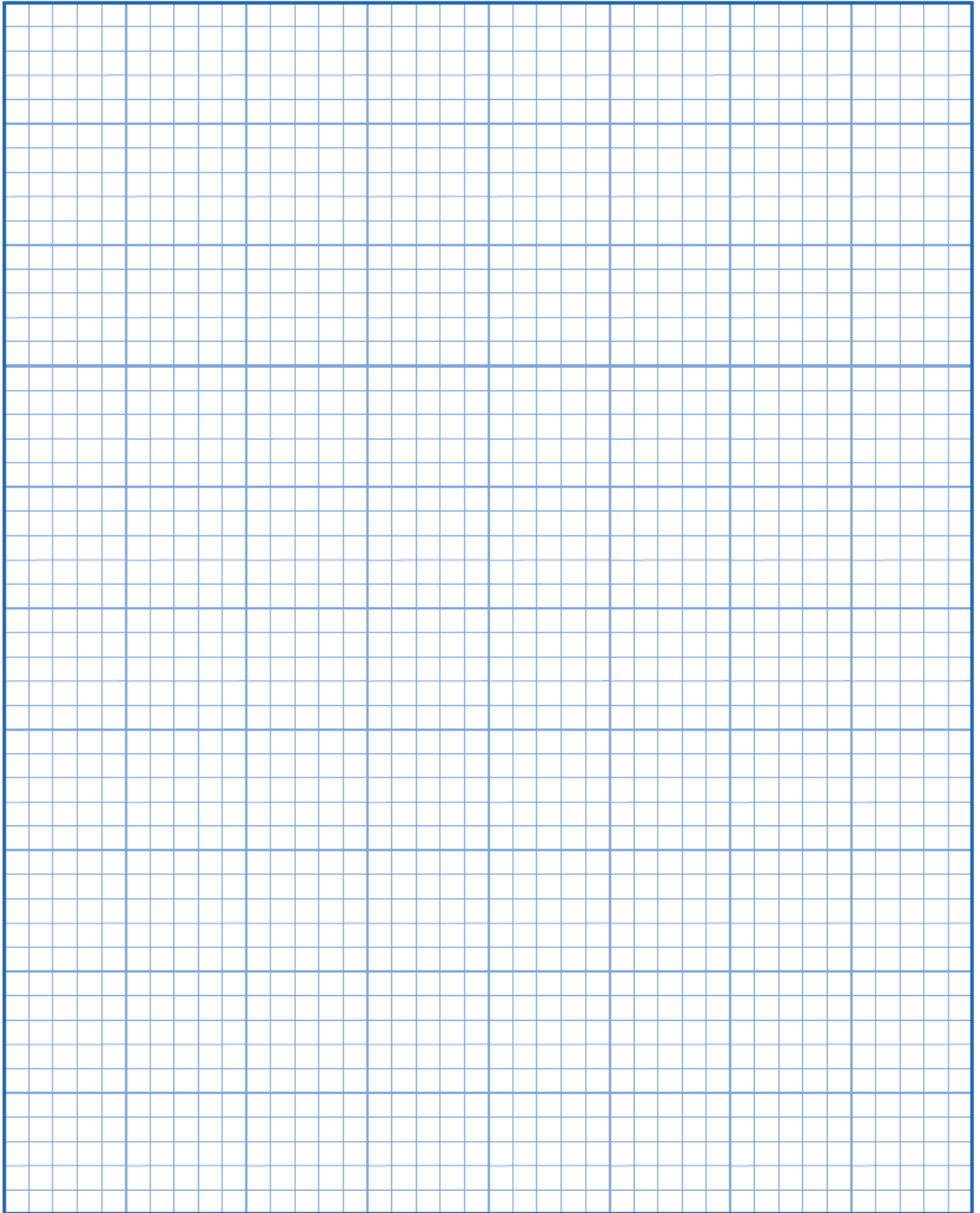


Working out the scale

First, look at your table of results and work out the largest number that has to go on each axis.

Now count the squares on your graph paper. If there are not enough, try again, counting in 2s, or 5s, or 10s, until you find a scale that fits on the paper.

· Graph: x-axis 'Force (N)' 0-7 (1 square = 1 N), y-axis 'Length (cm)' 0-45 (1 square = 5 cm); plot (0,5),(1,10),(2,15),(3,21),(4,26),(5,32),(6,36),(7,41) and join with a straight line of best fit.



Extension Task: Try to plot the results from this spring experiment on graph paper.

Two sets of results

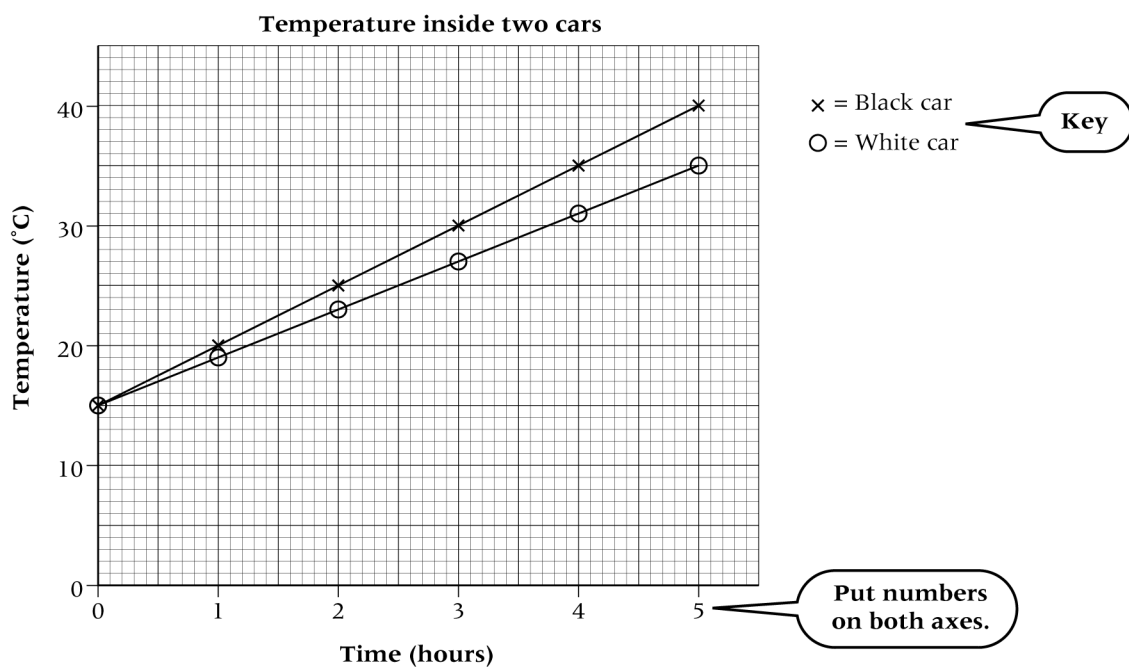
Sometimes you might want to put two sets of results on your graph so that you can compare them.

This table shows the temperature inside two cars that were parked in the sun.

Plot the points on the x-axis against time (in hours) on the y-axis, using the points from

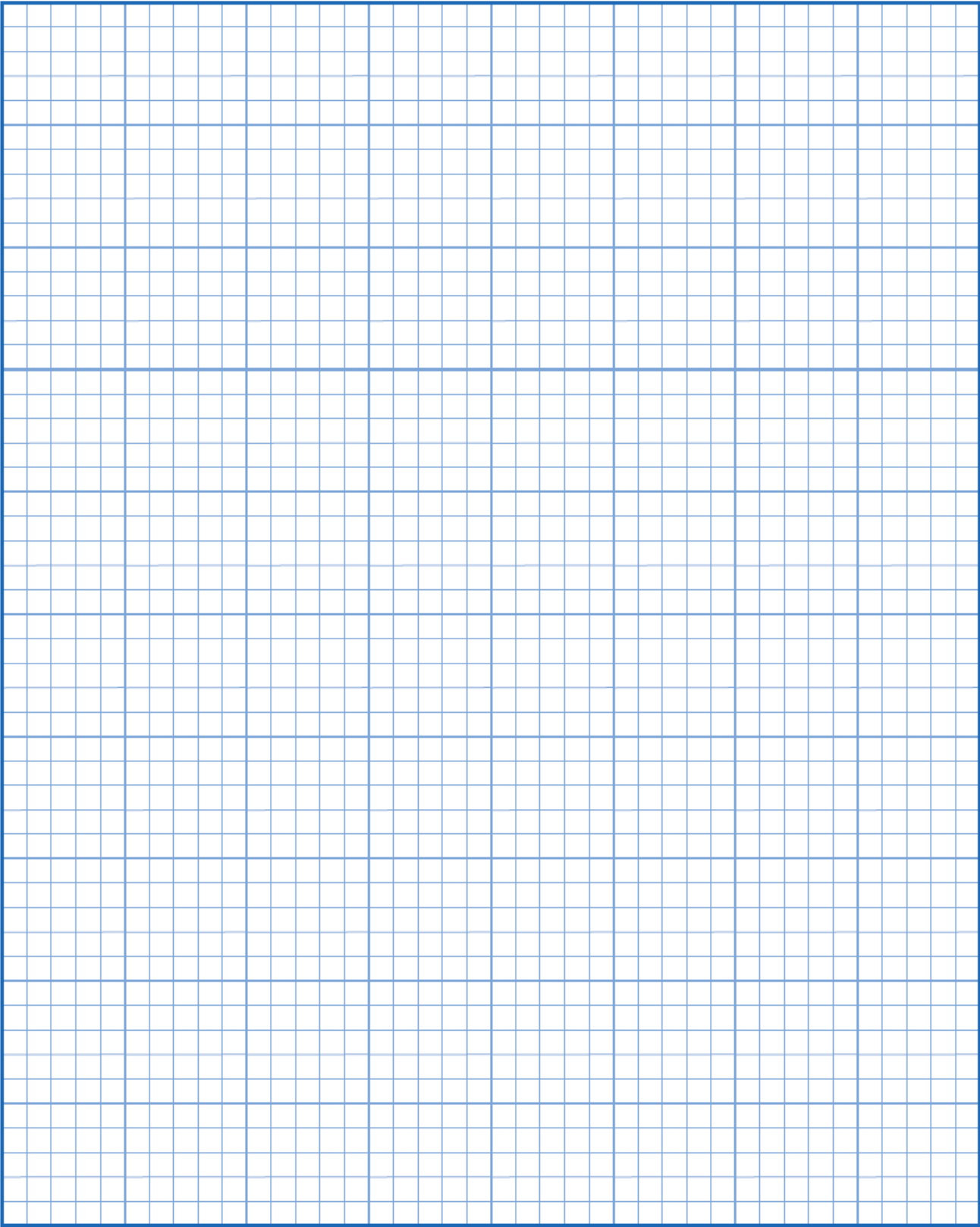
Time (hours)		Temperature (°C)			
Time (hours)		Black car		White car	
0	0	15	15	15	15
1	1	20	20	19	19
2	2	25	25	23	23
3	3	30	30	27	27
4	4	35	35	31	31
5	5	40	40	35	35

You can draw the two lines in different colours, or use different symbols for the points. Put a key on your graph.



- Plot a graph showing the temperatures inside the cars. Don't forget to put a title on your graph.

· Graph title 'Temperature inside cars parked in the sun'; x-axis 'Time (hours)' 0-5, y-axis 'Temperature (C)' 0-40; plot both sets of points, draw two lines and add a key.



7SS 6c Experiment Analysis

Name: _____

VERSION B1

Ms. Ellis notices that students who talk and are off-task during instruction tend to earn lower grades. She wonders if the amount of time spent off-task is related to overall grades. She hypothesizes that students who are off-task get lower grades. In order to test her hypothesis, she conducts the following experiment:

- Select 4 students who have the same I.Q.
- Provide all students with the same learning experiences.
- Provide all students with the same study materials.
- Observe student behavior and track time spent off-task daily.
- Compile overall grades and analyze the data.

The table shows the data gathered.

Student	Daily Time Off-task		Grade in class
1	0 minutes	0 minutes	95%
2	10 minutes	10 minutes	85%
3	20 minutes	20 minutes	70%
4	40 minutes	40 minutes	55%

- In this experiment, what is the independent variable? _____
- What is the dependent variable? The dependent variable is the grade in class (%), because it is what is measured and may be affected.
- List 3 control variables. Same I.Q., the same learning experiences and the same study materials.

Also the same teacher, the same lessons and the same tests for all four students.

- Create a line graph displaying the results. Make sure to:
 - Create an appropriate title and axis labels
 - Place the independent variable on the x-axis and the dependent variable on the y-axis
 - Calculate a reasonable scale for each axis
- The results show the independent and dependent variables have what type of relationship?

Circle one answer: direct relationship inverse relationship no relationship

- In a complete sentence, write a conclusion Ms. Ellis can draw after analyzing her results.

As the daily time spent off-task increased from 0 to 40 minutes the grade in class fell from 95% to 55%. so the

variable have an inverse relationship. 0 min = 70%, 40 min = 55%. The hypothesis that off-task students get lower grades is supported.

- Looking at the experiment design and results, list at least one "next step" related to the experiment.

Repeat the experiment with more students and over a longer time to check the pattern is the same.

Also record WHAT the off-task behaviour is, to see which types of off-task behaviour lower grades most.

• Line graph: title 'Time off-task and grade in class'; x-axis 'Daily time off-task (minutes)' 0-40, y-axis 'Grade (%)' 0-100; plot (0,95),(10,85),(20,70),(40,55) and join the points.

Name: _____

VERSION B2

Ms. Ellis notices that students who talk and are off-task during instruction tend to earn lower grades. She wonders if the amount of time spent off-task is related to overall grades. She hypothesizes that students who are off-task get lower grades. In order to test her hypothesis, she conducts the following experiment:

- Select 4 students who have the same I.Q.
- Provide all students with the same learning experiences.
- Provide all students with the same study materials.
- Observe student behavior and track time spent off-task daily.
- Compile overall grades and analyze the data.

The table shows the data gathered.

Student	Grade	Grade	Daily Time Off-task
1	90%	90%	0 minutes
2	70%	70%	10 minutes
3	80%	80%	20 minutes
4	55%	55%	35 minutes

- In this experiment, what is the independent variable?
The independent variable is the daily time spent off-task in minutes.
- What is the dependent variable?
The dependent variable is the grade in class (%).
- List 3 control variables. Same I.Q., the same learning experiences and the same study materials.
Also the same teacher, the same lessons and the same tests for all four students.
- Create a line graph displaying the results. Make sure to:
 - Create an appropriate title and axis labels
 - Place the independent variable on the x-axis and the dependent variable on the y-axis
 - Calculate a reasonable scale for each axis
- The results show the independent and dependent variables have what type of relationship?
Circle one answer: direct relationship inverse relationship no relationship
- In a complete sentence, write a conclusion Ms. Ellis can draw after analyzing her results.
Overall, students who spent more time off-task got lower grades (90% at 0 min falling to 55% at 35 min), so the hypothesis is supported.
Evidence: 0 min = 90%, 10 min = 70%, 35 min = 55%; student 3 (80% at 20 min) is the one result that does not fit the pattern.
- Looking at the experiment design and results, list at least one "next step" related to the experiment.
Test more students and repeat the lessons, so one odd result (student 3) cannot change the conclusion.
Track off-task time on more days and use the same recording method every time.

7SS 6d Experiment Analysis

Name: _____

VERSION C1

Mr. Reagan notices that students who read often also tend to be good writers. He wonders if time spent reading is related to higher writing scores. He hypothesizes that more time spent reading gives students more experience with story structure, vocabulary, spelling, and other elements of writing, which would improve their own writing abilities. In order to test his hypothesis, he conducts the following experiment:

- Conduct a reading/writing ability pretest and a reading habits survey with a 7th-grade class.
Select 4 students with the same ability score, but different reading habits.
- Provide all students with the same writing lessons throughout the year.
- Provide all students with the same writing assignments and amount of teacher feedback.
- Over the school year, keep track of student reading habits and scores on writing assessments.
- Compile overall grades and analyze the data.

The table shows the data gathered.

Student	Daily Time Spent Reading		Overall Writing Grade
1	10 minutes	10 minutes	58%
2	20 minutes	20 minutes	75%
3	35 minutes	35 minutes	86%
4	50 minutes	50 minutes	96%

- In this experiment, what is the independent variable? The daily time spent reading (minutes)
- What is the dependent variable? The overall writing grade (%)
- List 3 control variables.
Same reading/writing ability score at the start, same writing lessons, same writing assignments and amount of teacher feedback
teacher and the same length of study (one school year).
- Create a line graph displaying the results. Make sure to:
 - Create an appropriate title and axis labels
 - Place the independent variable on the x-axis and the dependent variable on the y-axis
 - Calculate a reasonable scale for each axis
- The results show the independent and dependent variables have what type of relationship?
Circle one answer: direct relationship inverse relationship no relationship
- In a complete sentence, write a conclusion Mr. Reagan can draw after analyzing her results.
As the daily time spent reading increased from 10 to 50 minutes the writing grade rose from 58% to 96%, showing a
direct relationship between the two variables. 10 min = 58%, 20 min = 75%, 35 min = 86%, 50 min = 96%. Mr. Reagan's hypothesis is supported.
- Looking at the experiment design and results, list at least one "next step" related to the experiment.
Repeat the study with more students and for longer to be sure the pattern is reliable.
Check that all students really did the same pretest, and measure the reading time with a reading log.

Name: _____

VERSION C2

Mr. Reagan notices that students who read often also tend to be good writers. He wonders if time spent reading is related to higher writing scores. He hypothesizes that more time spent reading gives students more experience with story structure, vocabulary, spelling, and other elements of writing, which would improve their own writing abilities. In order to test his hypothesis, he conducts the following experiment:

- Conduct a reading/writing ability pretest and a reading habits survey with a 7th-grade class.
Select 4 students with the same ability score, but different reading habits.
- Provide all students with the same writing lessons throughout the year.
- Provide all students with the same writing assignments and amount of teacher feedback.
- Over the school year, keep track of student reading habits and scores on writing assessments.
- Compile overall grades and analyze the data.

The table shows the data gathered.

Student	Overall Writing Grade		Daily Time Spent Reading
1	58%	58%	10 minutes
2	90%	90%	20 minutes
3	76%	76%	35 minutes
4	86%	86%	45 minutes

- In this experiment, what is the independent variable? The daily time spent reading (minutes)
- What is the dependent variable? The overall writing grade (%)
- List 3 control variables.
Same reading/writing ability score at the start, same writing lessons, same writing assignments and amount of teacher feedback
And the same length of study (one school year).
- Create a line graph displaying the results. Make sure to:
 - Create an appropriate title and axis labels
 - Place the independent variable on the x-axis and the dependent variable on the y-axis
 - Calculate a reasonable scale for each axis
- The results show the independent and dependent variables have what type of relationship?
Circle one answer: direct relationship inverse relationship no relationship
- In a complete sentence, write a conclusion Mr. Reagan can draw after analyzing her results.
On the whole, students who read for longer had higher writing grades (58% at 10 min rising to 86% at 45 min), so the variables have a direct relationship.
For example, 10 min = 58%, 20 min = 90%, 35 min = 76%, 45 min = 86%. This supports Mr. Reagan's hypothesis.
- Looking at the experiment design and results, list at least one "next step" related to the experiment.
Repeat with a bigger group and use the same reading log for everyone, because student 3 read more but scored lower.
Measure reading habits the same way each time so the results can be compared fairly.

7SS 6e Experiment Analysis

Name: _____

VERSION D1

Miguel notices his friends have skateboards of different lengths. He wonders if the length of the skateboard makes a difference in downhill speed. He hypothesizes that shorter boards will go faster than longer boards. In order to test his hypothesis, he conducts the following experiment:

- Select 4 skateboards of different lengths, but same widths
- Find a hill and mark a start and finish line.
- Ride each board down the hill using a stopwatch to record how long it takes from start to finish. Before start line, use only one small push off. During the ride, both feet will stay on the board at all times.
- Analyze the data.

The table shows the data gathered.

Trip #	Skateboard Length		Time on Hill
1	28 in.	28 in.	10 sec.
2	32 in.	32 in.	7 sec.
3	38 in.	38 in.	5 sec.
4	44 in.	44 in.	4 sec.

- In this experiment, what is the independent variable? The length of the skateboard (inches)
- What is the dependent variable? The time taken to ride down the hill (seconds)
- List 3 control variables. All boards are the same width, the same hill with the same start and finish lines, only one small push off before the start.
The same rider with both feet on the board, and the same stopwatch and timing method every trip.
- Create a line graph displaying the results. Make sure to:
 - Create an appropriate title and axis labels
 - Place the independent variable on the x-axis and the dependent variable on the y-axis
 - Calculate a reasonable scale for each axis
- The results show the independent and dependent variables have what type of relationship?
Circle one answer: direct relationship inverse relationship no relationship
- In a complete sentence, write a conclusion Miguel can draw after analyzing her results.
As the skateboard length increased from 28 to 44 inches the time on the hill fell from 10 to 4 seconds, so as length increased the time decreased (inverse relationship) Miguel's hypothesis that shorter boards go faster is not supported.
- Looking at the experiment design and results, list at least one "next step" related to the experiment.
Repeat every ride at least three times and average the times to make the results reliable.
Use the same rider for every trip and test more board lengths to see whether the trend continues.

Name: _____

VERSION D2

Miguel notices his friends have skateboards of different lengths. He wonders if the length of the skateboard makes a difference in downhill speed. He hypothesizes that shorter boards will go faster than longer boards. In order to test his hypothesis, he conducts the following experiment:

- Select 4 skateboards of different lengths, but same widths
- Find a hill and mark a start and finish line.
- Ride each board down the hill using a stopwatch to record how long it takes from start to finish. Before start line, use only one small push off. During the ride, both feet will stay on the board at all times.
- Analyze the data.

The table shows the data gathered.

Trip #	Time on Hill	Time on Hill	Skateboard Length
1	12 sec.	12 sec.	28 in.
2	10 sec.	10 sec.	32 in.
3	8 sec.	8 sec.	38 in.
4	10 sec.	10 sec.	44 in.

- In this experiment, what is the independent variable? The length of the skateboard (inches)
- What is the dependent variable? The time taken to ride down the hill (seconds)
- List 3 control variables. All boards are the same width, the same hill with the same start and finish lines, only one small push off before the start.
The same rider with both feet on the board, and the same stopwatch and timing method every trip.
- Create a line graph displaying the results. Make sure to:
 - Create an appropriate title and axis labels
 - Place the independent variable on the x-axis and the dependent variable on the y-axis
 - Calculate a reasonable scale for each axis
- The results show the independent and dependent variables have what type of relationship?
Circle one answer: direct relationship inverse relationship no relationship
- In a complete sentence, write a conclusion Miguel can draw after analyzing her results.
As the skateboard length increased the time on the hill generally got shorter (12 s to about 8-10 s), so the variables showed an inverse relationship.
So the hypothesis that shorter boards go faster is not supported; the 44 in. board (10 s) did not fit the pattern.
- Looking at the experiment design and results, list at least one "next step" related to the experiment.
Repeat each trip several times and average the times, because two boards produced the same 10 s result.
Test more board lengths with the same rider to see whether the time keeps getting shorter.

7SS 6f Experiment Analysis

Name: _____

VERSION E1

Sasha saw a YouTube video where Mentos candy caused Diet Coke to gush out of the bottle. She wonders if the number of Mentos used causes different kinds of "explosions". She hypothesizes that more Mentos will lead to bigger "explosions". In order to test her hypothesis, she conducts the following experiment:

- Set up four bottles of Diet Coke against a wall with height measurements marked off.
- One at a time, place a different amount of Mentos in each bottle.
- Set up video camera to help record the highest height achieved for each bottle.
- Analyze the data.

The table shows the data gathered.

Bottle	Height of explosion		# of Mentos
1	5 in.	5 in.	1
2	10 in.	10 in.	2
3	12 in.	12 in.	3
4	17 in.	17 in.	4

1. In this experiment, what is the independent variable? The number of Mentos added to the bottle

2. What is the dependent variable? The height of the explosion (inches)

3. ~~List 3 control variables~~ List 3 control variables: Diet Coke bottle, the same volume of Diet Coke and the same kind of Mentos each time.
The same temperature, the same wall with the height markings, and the same video camera and measuring method.

4. Create a line graph displaying the results. Make sure to:

- Create an appropriate title and axis labels
- Place the independent variable on the x-axis and the dependent variable on the y-axis
- Calculate a reasonable scale for each axis

5. The results show the independent and dependent variables have what type of relationship?
Circle one answer: direct relationship inverse relationship no relationship

6. In a complete sentence, write a conclusion Sasha can draw after analyzing her results.

As the number of Mentos increased from 1 to 4, the height of the explosion rose from 5 to 17 inches, so the variables have a direct relationship.

Evidence: 1 Mentos = 5 in., 2 = 10 in., 3 = 12 in., 4 = 17 in. Sasha's hypothesis that more Mentos give bigger explosions is supported.

7. Looking at the experiment design and results, list at least one "next step" related to the experiment.

Repeat each number of Mentos three times to check the results are reliable, and use the same bottle each time.

Test more amounts (for example 5, 6 and 7 Mentos) to find the biggest possible explosion.

Name: _____

VERSION E2

Sasha saw a YouTube video where Mentos candy caused Diet Coke to gush out of the bottle. She wonders if the number of Mentos used causes different kinds of "explosions". She hypothesizes that more Mentos will lead to bigger "explosions". In order to test her hypothesis, she conducts the following experiment:

- Set up four bottles of Diet Coke against a wall with height measurements marked off.
- One at a time, place a different amount of Mentos in each bottle.
- Set up video camera to help record the highest height achieved for each bottle.
- Analyze the data.

The table shows the data gathered.

Bottle	# of Mentos	# of Mentos	Height of explosion
1	2	2	10 in.
2	5	5	16 in.
3	10	10	8 in.
4	15	15	3 in.

- In this experiment, what is the independent variable? The number of Mentos added to the bottle
- What is the dependent variable? The height of the explosion (inches)
- List 3 control variables The same type and size of Diet Coke bottle, the same volume of Diet Coke and the same kind of Mentos each time.
The same temperature, the same wall with the height markings, and the same video camera and measuring method.
- Create a line graph displaying the results. Make sure to:
 - Create an appropriate title and axis labels
 - Place the independent variable on the x-axis and the dependent variable on the y-axis
 - Calculate a reasonable scale for each axis
- The results show the independent and dependent variables have what type of relationship?
Circle one answer: direct relationship inverse relationship no relationship
- In a complete sentence, write a conclusion Sasha can draw after analyzing her results.
The explosion was biggest with 5 Mentos (16 in.) and then fell as more Mentos were added (10 Mentos = 8 in., 15
Mentos = 3 in.), so overall more Mentos gave smaller explosions.
Mentos = 3 in.), so overall more Mentos gave smaller explosions and the hypothesis that more Mentos always give bigger explosions is not supported.
- Looking at the experiment design and results, list at least one "next step" related to the experiment.
Repeat each number of Mentos several times to check the pattern, especially between 1 and 5 Mentos.
Test amounts of 1, 3 and 5 Mentos to find the number that gives the biggest explosion.

7SS 6g Experiment Analysis

Name: _____

VERSION F1

Randy noticed that junk food nutrition labels have a lot of ingredients. He wonders if the number of ingredients is linked to healthfulness. He hypothesizes that foods with fewer ingredients are healthier. In order to test his hypothesis, he conducts the following experiment:

- Select four ice creams with different numbers of ingredients.
- Count the number of ingredients in each.
- Use the food app "Fooducate" to find an overall healthfulness score for each ice cream.
- Analyze the data.

The table shows the data gathered.

Ice-cream	# of ingredients	Healthfulness Score
1	5	80%
2	10	70%
3	12	65%
4	18	50%

- In this experiment, what is the independent variable? The number of ingredients in each ice cream
- What is the dependent variable? The healthfulness score (%)
- List 3 control variables. The same type of food (ice cream), the same serving size and the same flavour/brand type.
The same app (Fooducate) used in the same way, and the ingredient counts done on the same day.
- Create a line graph displaying the results. Make sure to:
 - Create an appropriate title and axis labels
 - Place the independent variable on the x-axis and the dependent variable on the y-axis
 - Calculate a reasonable scale for each axis
- The results show the independent and dependent variables have what type of relationship?
Circle one answer: direct relationship inverse relationship no relationship
- In a complete sentence, write a conclusion Randy can draw after analyzing her results.
As the number of ingredients increased from 5 to 18, the healthfulness score fell from 80% to 50%, so the variables have an inverse relationship. 10 = 70%, 12 = 65%, 18 = 50%. Randy's hypothesis that fewer ingredients are healthier is supported.
- Looking at the experiment design and results, list at least one "next step" related to the experiment.
Test more ice creams, including different brands, and repeat the Fooducate scores so the comparison is fairer.
Use the same serving size and check the labels yourself to be sure the ingredient counts are right.

Name: _____

VERSION F2

Randy noticed that junk food nutrition labels have a lot of ingredients. He wonders if the number of ingredients is linked to healthfulness. He hypothesizes that foods with fewer ingredients are healthier. In order to test his hypothesis, he conducts the following experiment:

- Select four ice creams with different numbers of ingredients.
- Count the number of ingredients in each.
- Use the food app "Fooducate" to find an overall healthfulness score for each ice cream.
- Analyze the data.

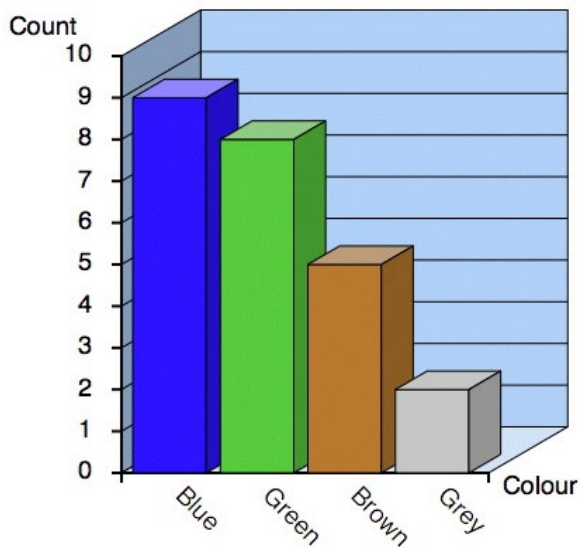
The table shows the data gathered.

Cereal	Healthfulness Score		% of ingredients
1	90%	90%	5
2	80%	80%	15
3	78%	78%	25
4	70%	70%	38

- In this experiment, what is the independent variable? The % (number) of ingredients in each cereal
- What is the dependent variable? The healthfulness score (%)
- List 3 control variables. The same type of food (cereal), the same serving size and the same kind of label information.
The same app (Fooducate) used in the same way, and all the scores found on the same day.
- Create a line graph displaying the results. Make sure to:
 - Create an appropriate title and axis labels
 - Place the independent variable on the x-axis and the dependent variable on the y-axis
 - Calculate a reasonable scale for each axis
- The results show the independent and dependent variables have what type of relationship?
Circle one answer: direct relationship inverse relationship no relationship
· Line graph: title '% of ingredients and healthfulness score'; x-axis '% of ingredients' 0-40, y-axis 'Healthfulness score (%)' 0-100; plot (5,90),(15,80),(25,78),(38,70).
- In a complete sentence, write a conclusion Randy can draw after analyzing her results.
As the % of ingredients increased from 5 to 38, the healthfulness score fell from 90% to 70%, so the variables have an inverse relationship.
The more ingredients a cereal contains, the lower its healthfulness score, so Randy's hypothesis that fewer ingredients are healthier is supported.
- Looking at the experiment design and results, list at least one "next step" related to the experiment.
Test more cereals and compare only foods of the same type and serving size.
Repeat the ingredient counts and the Fooducate scoring to check the results are reliable.

7SS Conclusion

Eye Colours in our Class



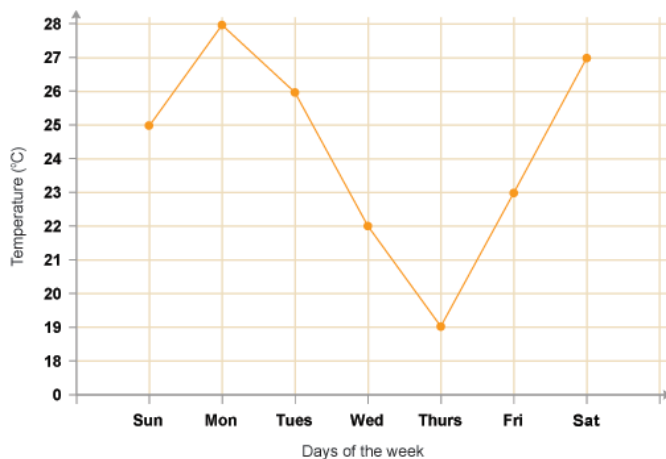
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Picture 1: use the 5 dotted lines to write the.....

conclusion -- the pattern found, evidence from.....

the results, why it happened and whether the.....

prediction was right.....



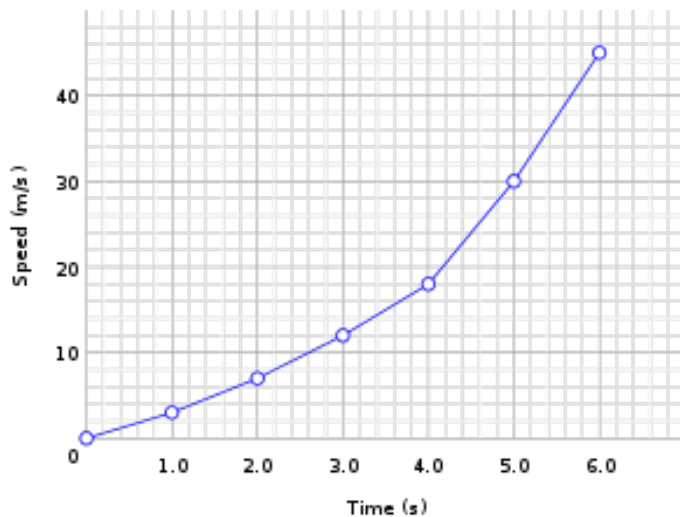
.....

Picture 2: (5 dotted lines): write the conclusion.....

for this investigation -- what happened, the.....

evidence from the data, the science explanation

and whether the hypothesis was supported.....



.....

Picture 3: (7 dotted lines): write a full conclusion

-- the results found, the evidence, why it.....

happened, and any improvement or next step.....

7SS 7a Growing Plants

Name: Date:

Student Exploration: Growing Plants

Directions: Follow the instructions to go through the simulation. Respond to the questions and prompts in the orange boxes.

Vocabulary: compost, fertilizer, mass, seed, soil, variable

Prior Knowledge Questions (Do these BEFORE using the Gizmo.)

1. What do you think plants need to grow and stay healthy?

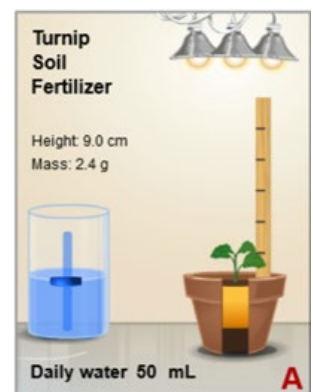
Plants need water, light, air (carbon dioxide), minerals/nutrients from the soil and a suitable temperature to grow and stay healthy.

2. **Soil** is a combination of tiny rock fragments and decayed plant materials. How do you think soil helps a plant? Soil holds the plant's roots in place and holds water and minerals, which the roots absorb, so the plant can grow and stay healthy.

Gizmo Warm-up: Grow the Biggest Plant!

1. In the Gizmo set up the three pots however you like:

- Choose a **seed** to drag into each pot.
- Click on the light bulbs to turn them on or off.
- Drag the **Water** slider up or down to set the amount of water each plant will get.
- If you like, drag **fertilizer** or **compost** into a pot.
- When the pots are ready, click **Play** (▶) and wait for the simulation to end.

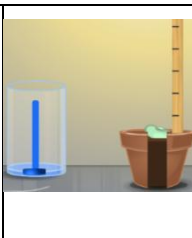


2. How tall was your tallest plant?

My tallest plant was about 90 cm tall (read the exact height from the ruler in the Gizmo).

3. Click **Reset** (↺) and **Clear pots**. Run a few more trials to grow the tallest plants you can. What conditions led to the tallest plant?

The tallest plants grew with two lights on, plenty of water (about 50 mL per day) and fertilizer or compost added to the soil.

Activity A: Wet and dry	<u>Get the Gizmo ready:</u> - Click Reset. - Click Clear pots.	
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Question: Do seeds need water to grow?

1. Form hypothesis: Do you think seeds need water to start growing? Explain why or why not.

Yes, I think seeds need water to start growing. Water is taken in by the seed so it swells, the seed coat splits and the young root and shoot can grow out.

2. Set up Gizmo: In the Gizmo, set up the three pots like this:

- Pot A: Bean seed, two lights on, 0 mL water per day
- Pot B: Bean seed, two lights on, 50 mL water per day
- Pot C: Bean seed, two lights on, 100 mL water per day

3. Experiment: Click **Play** to start. When the simulation is done, observe the plants.

4. Collect data: Fill in the data table below with the height and mass of each plant on day 50. (The **mass** of a plant is the amount of matter in the plant. It is related to how heavy it is.) In the last column describe what each plant looks like.

Pot	Water/day (mL)	Height (cm)	Mass (g)	Appearance
A	Pot A (0 mL/day): height 0 cm, mass 0 g - the bean did not grow at all, it stayed as a hard, dry seed.			
B	Pot B (50 mL/day): grew to about 60 cm with a mass of about 40 g - a tall, healthy, green plant with several leaves.			
C	Pot C (100 mL/day): grew into a healthy green plant, about the same size as Pot B (slightly shorter/lighter) - extra water did not make it bigger.			

5. Draw conclusions: Do seeds need water to grow? (Was your hypothesis correct?)

Yes - seeds need water to grow. Pot A had no water and did not grow, while Pots B and C both grew into healthy plants, so my hypothesis was correct.

6. Revise and repeat: Is more water always better? Create your own experiments to find the ideal amount of water for each kind of plant. Explain your findings below.

More water is not always better. About 50 mL per day gave the biggest plant; with 100 mL the plant was no bigger (the soil was too wet), so each plant has an ideal amount of water.

Activity B: Light and dark	<u>Get the Gizmo ready:</u> - Click Reset. - Click Clear pots.	
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Question: How does the amount of light affect how plants grow?

1. Form hypothesis: A **variable** is something that can be changed in an experiment, such as the amount of light. How do you think the amount of light affects how plants grow?

I think plants with more light will grow taller and heavier and the plant with no light will not grow at all, because light is needed for photosynthesis.

2. Experiment:

- Set up the three pots with 50 mL of water per day, plain soil, and turnip seeds.
- Turn on three lights over pot A, one light over pot B, and zero lights over pot C.
- Click **Play**.
- After 30 simulated days, click **Pause** (⏸).

3. Collect data: In the table below, describe each plant after 30 days.

Pot	Number of lights	Height (cm) on day 30	Mass (g) on day 30	Appearance on day 30
A	Pot A (3 lights): the tallest and heaviest plant - about 60 cm, dark green with many healthy leaves.			
B	Pot B (1 light): shorter and lighter than Pot A - about 40 cm, paler green with fewer leaves.			
C	Pot C (0 lights): almost no growth - only a few cm tall, pale yellow/white and weak, and it may die.			

4. Draw conclusions: Select the DATA tab. Look at the graphs for **Plant height** and **Plant mass**. How did the amount of light affect the growth of plants?

The more light the plants received the taller and heavier they grew (3 lights > 1 light > 0 lights). Light is needed for photosynthesis, so the plant in the dark could not make food and hardly grew.

5. Think and discuss: Click **Play** and let the simulation continue to 50 days. Based on what you have seen so far, what do seeds need to grow into healthy plants?

Seeds need water, air and a suitable temperature, and after germination they need light and minerals from the soil, to grow into healthy plants.

Activity C: Design an experiment	<u>Get the Gizmo ready:</u> - Click Reset. - Click Clear pots.	
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Question: You come up with the question! (See below.)

1. Create question: Fill in the blanks below with the variable and the type of plant you would like to study in this activity. (Do not repeat an experiment you have already done.)

How does the amount of fertilizer affect a bean plant?

2. Form hypothesis: What is your hypothesis for the question above? I think the bean plant grown with fertilizer will grow taller and heavier than the plant in plain soil, because fertilizer adds the minerals the plant needs to make new material.

3. Set up Gizmo: Set up the pots to test the variable you are investigating. Describe how you set up each pot in the table below.

Pot	Type of seed	Water/day	Number of lights	Type of soil
A				
B				
C				

Pot A: bean seed, 50 mL water/day, 2 lights, plain soil (no fertilizer) – this is the control pot.

Pot B: bean seed, 50 mL water/day, 2 lights, plain soil with fertilizer added.

4. Experiment: Click **Play** to start. When the simulation is done, observe the plants.
5. Collect data: Examine your plant data on the DATA tab. Record your results here.

Pot C: bean seed, 50 mL water/day, 2 lights, compost-rich soil.

Pot	Height (cm)	Mass (g)	Appearance
A	Pot A: about 45 cm, about 25 g - a healthy but smaller green plant (control).		
B	Pot B: about 65 cm, about 40 g - the tallest, heaviest plant with dark green leaves.		
C	Pot C: about 60 cm, about 35 g - a healthy green plant, slightly smaller than Pot B.		

6. Draw conclusions: What did you discover? Why do you think it happened that way?

The bean plants with fertilizer or compost grew taller and heavier than the plant in plain soil, because the extra minerals help the plant make more new material. Only the soil was changed.

Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p3: A research question is one sentence saying what you want to find out, e.g. 'How does the drop height affect the bounce height of a ball?'

p16: Read each measuring cylinder at the bottom of the meniscus at eye level and give the volume in cm³ (ml), e.g. a) 50 cm³, b) 20 cm³, c) 75 cm³, d) 30 cm³, e) 100 cm³.

p21: Diagram: beaker of water standing on a gauze on a tripod, thermometer in the water and the Bunsen burner underneath. Label Bunsen burner, air hole, tripod, gauze, beaker, water, thermometer.

p21: The roaring flame was the hottest. I know because it heated the same volume of water by the biggest temperature rise in the same time (the tallest bar on my graph).

p21: It was a fair test because only the independent variable (the type of flame) was changed; the volume of water, starting temperature, heating time and distance of the flame were all kept the same.

p21: Repeat each flame three times and average; use a digital thermometer; put a lid or insulation on the beaker to reduce heat loss; measure the distance of the flame below the gauze with a ruler.

p21: The hottest part is just above the tip of the inner blue cone. The gas is methane: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$. Sketch a Bunsen with the air hole open and label the hottest zone.

p21: Because the gas coming out of the top is not yet burnt - the inner cone is unburnt gas - so that part is the coolest and heating there is slow; the hottest zone is just above the inner cone.

p40: The independent variable is the daily time spent off-task in minutes, because it is the variable Ms. Ellis observes changing.

p41: Line graph: title 'Time off-task and grade'; x-axis 'Daily time off-task (minutes)' 0-35, y-axis 'Grade (%)' 0-100; plot (0,90),(10,70),(20,80),(35,55) and join the points.

p42: Line graph: title 'Reading time and writing grade'; x-axis 'Daily time spent reading (minutes)' 0-50, y-axis 'Writing grade (%)' 0-100; plot (10,58),(20,75),(35,86),(50,96) and join the points.

p43: Line graph: title 'Reading time and writing grade'; x-axis 'Daily time spent reading (minutes)' 0-45, y-axis 'Writing grade (%)' 0-100; plot (10,58),(20,90),(35,76),(45,86) and join the points.

p44: Line graph: title 'Skateboard length and time on hill'; x-axis 'Skateboard length (in.)' 0-45, y-axis 'Time on hill (sec.)' 0-12; plot (28,10),(32,7),(38,5),(44,4).

p45: Line graph: title 'Skateboard length and time on hill'; x-axis 'Skateboard length (in.)' 0-45, y-axis 'Time on hill (sec.)' 0-14; plot (28,12),(32,10),(38,8),(44,10).

p46: Line graph: title 'Number of Mentos and height of explosion'; x-axis '# of Mentos' 0-4, y-axis 'Height of explosion (in.)' 0-18; plot (1,5),(2,10),(3,12),(4,17).

p47: Line graph: title 'Number of Mentos and height of explosion'; x-axis '# of Mentos' 0-15, y-axis 'Height of explosion (in.)' 0-18; plot (2,10),(5,16),(10,8),(15,3).

p48: Line graph: title 'Number of ingredients and healthfulness score'; x-axis '# of ingredients' 0-20, y-axis 'Healthfulness score (%)' 0-100; plot (5,80),(10,70),(12,65),(18,50).